

The Field Surprisal Geometry — Full Worked Edition

Mahler–Gibbs Structure, Dual Flatness, Thermodynamics, Distances, and a Totally Geodesic Two–Statistic Surface

Echo–Squirrel Research (AceTheDactyl)

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Abstract

We attach a canonical positive–definite Fisher–Rao (surprisal) geometry to a number field’s emission catalog and develop it to four further, fully forced layers. The catalog’s Mahler measures $\{2, 3, 5, \varphi, \varphi, \varphi^4, \varphi^4-1\}$ sum to $Z = 17+4\sqrt{5}$ and, through the description cost $\log M$, define the *Mahler–Gibbs family* $p_i(\beta) \propto M_i^{-\beta}$, a forced exponential family. On it we prove: surprisal is affine in cost with slope β and Fisher equals the cost variance (§4); the geometry is *dually flat*, with the Gibbs curve an e –geodesic and KL between temperatures a Bregman divergence of $\log Z$ (§5); its thermodynamics obey the identity $C(\beta) = \beta^2 I(\beta)$ —Fisher information is a heat capacity—with a distinguished peak–fluctuation temperature (§6); its Fisher–Rao distances are Bhattacharyya angles, the seven seeds forming a regular spherical simplex at mutual distance π , with an explicit information length (§7). Adjoining the degree as a second forced statistic yields a two–parameter field geometry whose Fisher matrix is $\nabla^2 A$; we prove its Gaussian curvature is *exactly* the constant $\frac{1}{4}$ —an isometric round $\frac{1}{4}$ –sphere, embedded as a *ruled* surface (swept by great circles)—because a single–outcome indicator makes the family a spherical suspension whose metric is a round sphere for any curve (§10). No such surface is totally geodesic: $\log M$ alone spans a 6–dimensional algebra, so none lies in a great subsphere. Generic and multi–set–indicator statistics are not even constant curvature (controls give $4/21$, $857/16928$, and sign–changing values), and among the catalog’s forced invariants only degree—the single–outcome indicator—gives constant $\frac{1}{4}$ (§11), joined by a second over φ^4 from the conjugate–angle charge; in fact the constant–curvature– $\frac{1}{4}$ surfaces are *exactly eight*—one per seed plus the golden–pair merge—and no non–indicator statistic gives another (§12). Two independent catalogs form a Riemannian product with block–diagonal Fisher and zero cross–curvature (§13). Every load–bearing identity is machine–verified (`field_surprisal_v2.py`, `suspension_theorem.py`, `field_surprisal_tier2.py`; 12/12, 9/9, 15/15; exact arithmetic). This edition closes four fronts. The eight–surface classification is proved *in full* by a finite branch calculus—an exact window identity plus a plane lemma (§13)—so the non–indicator exclusion is now [FORCED] with no sampling residue. The temperature is dissolved: product functoriality plus measurable Cauchy *forces* $p \propto M^{-\beta}$, making β a coordinate on a forced e –geodesic of finite total information length $L_{\text{tot}} \approx 5.6462$ running from the golden merge to the φ^4 vertex (§15). Iterated indicator joins give round $\frac{1}{4}$ –spheres in every dimension, realized by the catalog triple (cost, degree, charge parity) (§16); and the first totally geodesic family appears at exactly $k = 5$, uniquely the golden–merged simplex $\mathbb{R}[\log M]$. A fifth front closes in this revision: the window identity is proved *conceptually*— $P = Z^2 \sum_{|s|=4} q_4(s) w_s$ by Sylvester’s determinant identity plus Cauchy–Binet (Theorem 13.4)—resolving Open Problem (3) and retiring the branch calculus from the critical path. This revision then closes three of the remaining open fronts: the master identity lifts to every k (Sylvester + Cauchy–Binet over $(k+2)$ –subsets), yielding windowed Gauss obstructions, a rank– ≤ 1 window criterion for constant $\frac{1}{4}$, and a complete $k = 3$ double–indicator census—exactly 26 windowwise–flat classes, the disjoint within–level pairs; the eight–surface census is proved *catalog–invariant*, with count $\sum_{\ell} (2^{m_{\ell}} - 1)$ over cost levels (generic catalogs need no

branch analysis at all, and a new branch–rank calculus closes necessity for four catalogs including the full one); and the temperature selection resolves as a *dichotomy*—no metric–canonical operating point exists, and the candidate principles provably disagree, so the declared election is the answer’s final form. This revision closes the last two structural fronts. Necessity holds at *every* k : constant $\frac{1}{4}$ is exactly windowwise rank ≤ 1 , because window collisions are squarefree—the Salem square cannot act on windows, only the golden swap can, and twin symmetry neutralizes it—so the 26–class census is the complete constant– $\frac{1}{4}$ double–indicator landscape and the analytic problem reduces to finite linear algebra. And the compositum *couples*: on its own tensor catalog the Mahler–Gibbs cost is provably non–separable (interaction contrast exactly $\log 2$; golden sector $-6 \log \varphi$; interaction tensor of exact rank 5), resolving the multi–field question on its forced horn. This final revision completes the program: the landscape is classified at every k —constant $\frac{1}{4}$ is exactly the *partitioned–affine* condition, by a moment trichotomy on window circuits, with catalog counts 8/56/95/31/1 and the indicator–join conjecture refuted—and the coupled compositum family is proved *not* constant– $\frac{1}{4}$: the coupling is curvature–visible, already at the uniform point where the cross–covariance vanishes exactly, and Δ is not determined by the charge data. Sixteen harnesses (61/61, 17/17, 4/4, 16/16, 12/12, 24/24, 44/44, 16/16, 13/13, 9/9, 72/72, 18/18, 28/28, 13/13, 21/21, 9/9; exact arithmetic over $\mathbb{Q}(\log 2, \log 3, \log 5, \log \varphi)$, plus $\log 7$ for the extended catalog) verify every new decision.

1 Introduction and discipline

The companion $\lambda = 2c$ manuscript charges a number field’s growth rule a description cost $\log M(\theta)$ and reads its parameter geometry as the (flat) trace form. Here we build the complementary object: the positive–definite Fisher–Rao geometry of the catalog’s outcome distribution, and four forced layers on top of it. Nothing new is assumed about the field; we use only its forced data. Each claim is tagged [FORCED] (theorem/exact computation), [COMPUTED] (validated numeric), [DECLARED] (a principled invariance–free choice), or [OPEN]. All symbolic identities are checked in exact **sympy**; the one numeric object (§10 curvature) uses a routine validated against four metrics of known curvature.

2 The catalog and its forced Mahler data

seed	minimal polynomial	M (exact)	degree
$\sqrt{2}$	$x^2 - 2$	2	2
$\sqrt{3}$	$x^2 - 3$	3	2
$\sqrt{5}$	$x^2 - 5$	5	2
φ	$x^2 - x - 1$	$\varphi = (1 + \sqrt{5})/2$	2
τ	$x^2 + x - 1$	φ	2
φ^4	$x^2 - 7x + 1$	$\varphi^4 = (7 + 3\sqrt{5})/2$	2
K	$x^4 + 5x^2 - 5$	$\varphi^4 - 1 = (5 + 3\sqrt{5})/2$	4

Table 1: The catalog Ω , with $M(\alpha) = \prod_{|\zeta|>1} |\zeta|$ and degrees. All exact and machine–verified.

Each measure is $\prod_{|\zeta|>1} |\zeta|$ over the roots ζ . For a quadratic $x^2 - D$ ($D > 1$) both roots $\pm\sqrt{D}$ lie outside the unit circle, giving $M = D$; the golden seeds keep only the dominant root, $M(\varphi) = M(\tau) = \varphi$, and $M(\varphi^4) = \varphi^4$. The quartic K is the one non–obvious case: $x^4 + 5x^2 - 5 = 0$

gives $x^2 = \frac{-5 \pm 3\sqrt{5}}{2}$, hence real roots $\pm K$ with $|K| = 5^{1/4}/\varphi < 1$ (inside) and imaginary roots $\pm i\beta$ with $\beta^2 = \frac{5+3\sqrt{5}}{2} > 1$ (outside); only the imaginary pair contributes, so $M(K) = \beta^2 = \varphi^4 - 1$.

Lemma 2.1 (Field constant). $Z := \sum_{\alpha \in \Omega} M(\alpha) = 10 + 2\varphi + 2\varphi^4 - 1 = 17 + 4\sqrt{5}$. [FORCED]

Proof. $2 + 3 + 5 = 10$; $\varphi + \varphi = 1 + \sqrt{5}$; $\varphi^4 + (\varphi^4 - 1) = 2 \cdot \frac{7+3\sqrt{5}}{2} - 1 = 6 + 3\sqrt{5}$; summing gives $10 + (1 + \sqrt{5}) + (6 + 3\sqrt{5}) = 17 + 4\sqrt{5}$. $\square$

The integer 17 is a computed coincidence of this catalog, not a structural invariant, and is read as nothing more.

3 The Mahler–Gibbs family

The forced cost is $E_i := \log M_i$; the maximum–entropy law at inverse temperature β is

$$p_i(\beta) = \frac{M_i^{-\beta}}{Z(\beta)}, \quad Z(\beta) = \sum_j M_j^{-\beta}, \quad A(\beta) := \log Z(\beta).$$

This is an exponential family with sufficient statistic $\log M$ (structure [FORCED]; the value of β is [DECLARED]). Distinguished temperatures: $\beta = 0$ (uniform); $\beta = 1$ (MDL/Occam prior $p \propto 1/M$); $\beta = \lambda = \sqrt{5}$ (the field’s forced exchange rate); $\beta = -1$ (exact anchor $p \propto M$, algebraic in $\mathbb{Q}(\sqrt{5})$).

4 Surprisal geometry

Proposition 4.1 (Surprisal–cost affinity). $S_i(\beta) := -\log p_i(\beta) = \beta \log M_i + \log Z(\beta)$; hence $\partial S_i / \partial (\log M_i) = \beta$ and $S_i - S_j = \beta \log(M_i/M_j)$. [FORCED]

Proof. From $p_i(\beta) = M_i^{-\beta} / Z(\beta)$, $S_i = -\log p_i = -\log M_i^{-\beta} + \log Z(\beta) = \beta \log M_i + \log Z(\beta)$. The partial derivative and the difference are immediate. The slope of surprisal against the Mahler cost is exactly the inverse temperature β ; at $\beta = \lambda = \sqrt{5}$ the surprisal therefore equals the growth–rule cost $\lambda \log M_i$ up to the common constant $\log Z$. $\square$

Proposition 4.2 (Fisher = cost variance). $I(\beta) = A''(\beta) = \text{Var}_\beta(\log M) \geq 0$, where $A(\beta) := \log Z(\beta)$. [FORCED]

Proof. Since $Z'(\beta) = \sum_j M_j^{-\beta} (-\log M_j)$,

$$A'(\beta) = \frac{Z'(\beta)}{Z(\beta)} = -\sum_j p_j(\beta) \log M_j = -\mathbb{E}_\beta[\log M].$$

Differentiating once more,

$$A''(\beta) = -\frac{d}{d\beta} \mathbb{E}_\beta[\log M] = \mathbb{E}_\beta[(\log M)^2] - (\mathbb{E}_\beta[\log M])^2 = \text{Var}_\beta(\log M),$$

the standard identity that the second derivative of the log–partition of an exponential family is the variance of its sufficient statistic; non–negativity is that of a variance. $\square$

Lemma 4.3 (The geometry is a sphere). The Fisher–Rao metric $g_{ij} = \delta_{ij}/p_i$ on the interior of Δ^{m-1} is isometric, via $\Phi(p) = 2\sqrt{p}$, to the positive orthant of the sphere $S^{m-1}(2)$; its constant sectional curvature is $\frac{1}{4}$. [FORCED]

Proof. With $\Phi(p) = (2\sqrt{p_1}, \dots, 2\sqrt{p_m})$ one has $d(2\sqrt{p_i}) = p_i^{-1/2} dp_i$, so the pullback of the Euclidean metric is $\sum_i (d(2\sqrt{p_i}))^2 = \sum_i p_i^{-1} dp_i^2$, which is exactly g . The image satisfies $\sum_i (2\sqrt{p_i})^2 = 4 \sum_i p_i = 4$, hence lies on the sphere of radius 2; a round sphere of radius r has constant sectional curvature $1/r^2$, here $\frac{1}{4}$ (Rao [2]; the $m = 3$ case is checked symbolically in Appendix A). The Gibbs curve $\beta \mapsto p(\beta)$ thus lives on a fixed curvature $\frac{1}{4}$ sphere, and its length element is $\sqrt{I(\beta)} = \sqrt{\text{Var}_\beta(\log M)}$. $\square$

5 Dual flatness

The family is a dually flat manifold in the sense of Amari–Nagaoka [1]: the natural (e -affine) coordinate is β , the expectation (m -affine) coordinate is $\eta = A'(\beta) = -U(\beta)$ with $U(\beta) := \mathbb{E}_\beta[\log M]$ the mean cost, and the two dual potentials are $A(\beta) = \log Z$ and its Legendre transform, the negentropy $A^*(\eta) = -H(p)$.

Proposition 5.1 (Bregman form of KL). *The Kullback–Leibler divergence between two temperatures is the Bregman divergence of $\log Z$:*

$$\text{KL}(p(\beta_1) \| p(\beta_2)) = A(\beta_2) - A(\beta_1) + (\beta_2 - \beta_1) U(\beta_1).$$

In particular the Gibbs curve $\beta \mapsto p(\beta)$ is an e -geodesic (a straight line in the natural coordinate), and $A^(\eta) = -H$ holds identically.* [FORCED]

Proof. For any i ,

$$\log \frac{p_i(\beta_1)}{p_i(\beta_2)} = (-\beta_1 \log M_i - A(\beta_1)) - (-\beta_2 \log M_i - A(\beta_2)) = (\beta_2 - \beta_1) \log M_i + A(\beta_2) - A(\beta_1).$$

Averaging over $p(\beta_1)$ and using $\mathbb{E}_{\beta_1}[\log M] = U(\beta_1)$,

$$\text{KL}(p(\beta_1) \| p(\beta_2)) = (\beta_2 - \beta_1) U(\beta_1) + A(\beta_2) - A(\beta_1),$$

which is the Bregman divergence generated by the convex potential A . The Legendre identity $A^*(\eta) = \beta\eta - A(\beta) = \sum_i p_i \log p_i = -H$ is the standard exponential-family duality. Both are verified in Appendix A. $\square$

As a worked instance, taking $\beta_1 = -1$, $\beta_2 = 1$ and reading A, U from the thermodynamic table of §6, $\text{KL}(p(-1) \| p(1)) = A(1) - A(-1) + 2U(-1) = 0.950 - 3.256 + 2(1.458) = 0.610$ nats—a description-length gap of 0.610 nats between the anchor and the MDL prior.

Consequently the generalized Pythagorean theorem holds: if the e -geodesic from p to q meets the m -geodesic from q to r Fisher-orthogonally, then $\text{KL}(p \| r) = \text{KL}(p \| q) + \text{KL}(q \| r)$.

6 Thermodynamics of the catalog

With energies $E_i = \log M_i$ and temperature $T = 1/\beta$: free energy $F(\beta) = -A(\beta)/\beta$, internal energy $U(\beta) = -A'(\beta)$, and Gibbs entropy $S(\beta) = A(\beta) + \beta U(\beta)$, which coincides with the Shannon entropy $H(p(\beta))$.

Proposition 6.1 (Fisher information is a heat capacity).

$$C(\beta) = \frac{dU}{dT} = \beta^2 I(\beta) = \beta^2 \text{Var}_\beta(\log M) \geq 0.$$

[FORCED]

Proof. $dU/d\beta = -A''(\beta) = -I(\beta)$ and $dT/d\beta = -1/\beta^2$, so $C = (dU/d\beta)/(dT/d\beta) = \beta^2 I(\beta)$; the non-negativity is $I \geq 0$ (verified in Appendix A). $\square$

The fluctuation $\text{Var}_\beta(\log M)$ peaks at a distinguished *peak-fluctuation temperature* $\beta^* \approx -0.0768$, where $\text{Var}_{\beta^*}(\log M) \approx 0.3289$ [COMPUTED]; the heat capacity $C = \beta^2 I$ is however suppressed near $\beta = 0$ by the β^2 factor and grows with $|\beta|$. As $\beta \rightarrow +\infty$ the measure concentrates on the minimal-Mahler seeds φ, τ , and as $\beta \rightarrow -\infty$ on the maximal-Mahler seed φ^4 . Table 2 records the functions at the distinguished temperatures.

β	$Z(\beta)$	$A = \log Z$	$U = \langle \log M \rangle$	$S = H$	$C = \beta^2 I$	$F = -A/\beta$
-1	25.944	3.256	1.458	1.798	0.2626	3.256
$\beta^* \approx -0.077$	7.654	2.035	1.176	1.945	0.0019	26.51
0	7	1.946	1.151	1.946	0	—
1	2.586	0.950	0.855	1.806	0.243	-0.950
$\sqrt{5}$	1.040	0.039	0.648	1.487	0.520	-0.018

Table 2: Thermodynamics of the Mahler–Gibbs family at the distinguished temperatures (A, U, S, C in nats). Entropy $S = H$ is maximal, $\log 7 \approx 1.946$, at $\beta = 0$; the fluctuation peaks at β^* ; the heat capacity $C = \beta^2 I$ vanishes at $\beta = 0$ and grows on either side. All values [COMPUTED].

7 Distances and information length

Proposition 7.1 (Fisher–Rao distance). *For distributions p, q , the geodesic distance is the Bhattacharyya angle $d_{\text{FR}}(p, q) = 2 \arccos \sum_i \sqrt{p_i q_i}$. The seven pure states (vertices) are mutually equidistant, $d_{\text{FR}}(\delta_i, \delta_j) = \pi$ for $i \neq j$, forming a regular spherical simplex.* [FORCED]

Proof. Under $\Phi(p) = 2\sqrt{p}$ (Lemma 4.3) the images $\Phi(p), \Phi(q)$ lie on the radius-2 sphere with Euclidean inner product $\Phi(p) \cdot \Phi(q) = 4 \sum_i \sqrt{p_i q_i}$; on a sphere of radius r the geodesic distance is r times the central angle, here $2 \arccos(\frac{1}{4} \Phi(p) \cdot \Phi(q)) = 2 \arccos \sum_i \sqrt{p_i q_i}$. For distinct vertices the Bhattacharyya coefficient $\sum_k \sqrt{\delta_i(k) \delta_j(k)} = 0$, so $d_{\text{FR}} = 2 \arccos 0 = \pi$. $\square$

The information length of the Gibbs curve is $L(\beta_0, \beta_1) = \int_{\beta_0}^{\beta_1} \sqrt{I(\beta)} d\beta = \int \sqrt{\text{Var}_\beta(\log M)} d\beta$. Between the exact anchor and the exchange-rate temperature, $L(-1 \rightarrow \sqrt{5}) \approx 1.5999$ nats, while the chord distance $d_{\text{FR}}(p(-1), p(\sqrt{5})) \approx 1.5812 \leq L$ [COMPUTED], as an arc bounds its chord. Table 3 gives the full distance matrix among the distinguished temperatures.

8 The exact anchor ($\beta = -1, p \propto M$)

Here $p_i = M_i/Z$, $Z = 17 + 4\sqrt{5}$, all algebraic in $\mathbb{Q}(\sqrt{5})$, and the metric is the exact diagonal $g = \text{diag}(Z/M_i)$.

9 The keystone–power sub-model: entry and order

With $R = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, $R^n = F_n R + F_{n-1} I$ and $M(\text{charpoly } R^n) = \varphi^n$, so on the ordered sequence $\{R^n : n_0 \leq n \leq N\}$ the cost is linear, $E_n = n \log \varphi$.

d_{FR}	$\beta = -1$	$\beta = 0$	$\beta = 1$	$\beta = \sqrt{5}$
$\beta = -1$	0	0.553	1.092	1.581
$\beta = 0$	0.553	0	0.542	1.040
$\beta = 1$	1.092	0.542	0	0.502
$\beta = \sqrt{5}$	1.581	1.040	0.502	0

Table 3: Fisher–Rao distances $d_{\text{FR}} = 2 \arccos \sum_i \sqrt{p_i q_i}$ (nats) between the Gibbs distributions at the distinguished temperatures. Distances are monotone along the ordered curve but *sub-additive*: $d(-1, \sqrt{5}) = 1.581 < 0.553 + 0.542 + 0.502 = 1.597$. This is the signature that the Gibbs curve is an e -geodesic (straight in β , §5) but *not* a Fisher–Rao metric geodesic—the metric geodesic between the endpoints is shorter, and the full arc length 1.600 (the information length) exceeds the chord 1.581. Every pure-state pair sits at the maximal distance π . All values [COMPUTED].

seed	$p_i = M_i/Z$	p_i	$S_i = -\log p_i$
$\sqrt{2}$	$2/(17 + 4\sqrt{5})$	0.0771	2.5628
$\sqrt{3}$	$3/(17 + 4\sqrt{5})$	0.1156	2.1573
$\sqrt{5}$	$5/(17 + 4\sqrt{5})$	0.1927	1.6465
φ	$\varphi/(17 + 4\sqrt{5})$	0.0624	2.7747
τ	$\varphi/(17 + 4\sqrt{5})$	0.0624	2.7747
φ^4	$\varphi^4/(17 + 4\sqrt{5})$	0.2642	1.3311
K	$(\varphi^4 - 1)/(17 + 4\sqrt{5})$	0.2256	1.4888

Table 4: The exact anchor distribution ($\beta = -1$, $p \propto M$): weights exact in $\mathbb{Q}(\sqrt{5})$, surprisals for display. The measure favours the high–Mahler seeds φ^4, K and most suppresses the golden pair φ, τ (equal weight, as $M(\varphi) = M(\tau)$). The seven weights sum to 1.

Proposition 9.1 (Geometric sub-model). *With $r = \varphi^{-\beta} \in (0, 1)$, the Gibbs law is the shifted geometric $p_n = (1-r)r^{n-n_0}$, $Z = \frac{r^{n_0} - r^{N+1}}{1-r} \rightarrow \frac{r^{n_0}}{1-r}$, the surprisal is linear $S_n = (n-n_0)\beta \log \varphi - \log(1-r)$, and $I(\beta) = (\log \varphi)^2 r/(1-r)^2$. The point of entry n_0 shifts the law; the order n is its support.* [FORCED]

The sequence data are exact integers and their forced measures (Table 5); the row $n = 4$ carries the catalog gap seed $x^2 - 7x + 1$, whose trace $L_4 = 7$ is the normalisation appearing throughout the corpus.

10 Two-statistic field geometry: a constant-curvature ruled surface

Adjoin the degree d_i as a second forced statistic; note $d_i = 2 + 2 \cdot \mathbf{1}_{\{i=K\}}$, so degree is affinely the indicator of the single outcome K . The two-parameter family $p_i \propto \exp(-\beta_1 \log M_i - \beta_2 d_i)$ is an exponential family whose Fisher matrix is the Hessian $\nabla^2 A(\beta_1, \beta_2) = \text{Cov}_\beta(\log M, \text{deg})$. At the uniform point,

$$g = \begin{pmatrix} 0.3283 & 0.1761 \\ 0.1761 & 0.4898 \end{pmatrix}, \quad \text{Var}(\text{deg}) = \frac{24}{49} = 0.4898,$$

n	F_n	$L_n = \varphi^n + \psi^n$	$M(R^n) = \varphi^n$	cost $E_n = n \log \varphi$
1	1	1	1.618	0.481
2	1	3	2.618	0.962
3	2	4	4.236	1.444
4	3	7	6.854	1.925
5	5	11	11.090	2.406
6	8	18	17.944	2.887

Table 5: The keystone–power sub–catalog $\{R^n\}$: Fibonacci F_n , Lucas L_n , the forced Mahler measure φ^n , and the linear cost $n \log \varphi$. Linearity of the cost is what makes the Gibbs law exactly geometric (Prop. 9.1).

symmetric positive definite. This is the genuinely two–dimensional field metric; we now determine its curvature exactly.

Theorem 10.1 (Indicator–suspension). *Let (a, c) be statistics on a finite set Ω with positive base measure ν , and suppose c is affinely the indicator of a single outcome K (i.e. $c = \alpha + \beta \mathbf{1}_K$, $\beta \neq 0$) while a is non–constant on $\Omega \setminus \{K\}$. Then the Fisher–Rao geometry of $p_i \propto \nu_i e^{\theta_1 a_i + \theta_2 c_i}$ has constant Gaussian curvature $\frac{1}{4}$; it is an isometric round $\frac{1}{4}$ –sphere, embedded in $S^{|\Omega|-1}(2)$ as a ruled surface swept by great circles (the c –direction flow lines). It is not totally geodesic unless a is affinely constant on $\Omega \setminus \{K\}$ (which fails whenever there are three or more distinct a –values). [FORCED]*

Proof. The Fisher metric of an exponential family depends on the statistics only through the affine span $\langle 1, a, c \rangle$ and is invariant, as a Riemannian manifold, under affine reparametrization; so replace c by $\mathbf{1}_K$ and parametrize by (θ, q) , with $q = p_K \in (0, 1)$ the apex mass and θ the natural parameter of the conditional family $\rho(\theta)$ on $\Omega \setminus \{K\}$, so $p = ((1 - q)\rho(\theta), q)$. Pass to square–root coordinates: $p \mapsto \sqrt{p}$ is an isometry onto the unit sphere whose round metric is $\frac{1}{4}$ the Fisher metric. With $x(\theta) = \sqrt{\rho(\theta)}$ a unit vector ($|x|^2 = \sum_i \rho_i = 1$) tracing a curve on the equatorial sphere, and $\sqrt{q} = \sin \psi$ ($\psi \in (0, \frac{\pi}{2})$),

$$\sqrt{p} = (\cos \psi x(\theta), \sin \psi), \quad \partial_\psi \sqrt{p} = (-\sin \psi x, \cos \psi), \quad \partial_\theta \sqrt{p} = (\cos \psi x', 0).$$

Using $|x|^2 = 1 \Rightarrow x \cdot x' = 0$, the induced unit–sphere metric is

$$g^{\text{unit}} = \underbrace{(\sin^2 \psi |x|^2 + \cos^2 \psi)}_{=1} d\psi^2 + 2 \underbrace{(-\sin \psi \cos \psi x \cdot x')}_{=0} d\psi d\theta + \cos^2 \psi |x'|^2 d\theta^2 = d\psi^2 + \cos^2 \psi d\ell^2,$$

reparametrizing θ by the arc length ℓ of x ($d\ell = |x'| d\theta$). This is the round unit–sphere metric (latitude ψ , longitude ℓ), of Gaussian curvature $-f''/f = 1$ with $f = \cos \psi$ —independent of the curve x . As the Fisher metric is $4g^{\text{unit}}$, its curvature is the constant $\frac{1}{4}$. The family is thus the spherical suspension of the apex K over the a –curve; non–constancy of a makes $x(\theta)$ a genuine immersed curve, so the surface is two–dimensional. $\square$

The embedding structure is exactly a ruled surface. Writing $s = \sqrt{p}$ and $\tau_\alpha = T_\alpha - \langle T_\alpha \rangle$ (centred statistics), the second fundamental form in the sphere is $\Pi_{\alpha\beta} = \frac{1}{4}(\tau_\alpha \tau_\beta s)^\perp$, and $\Pi_{\alpha\beta} = 0$ iff the pointwise product $T_\alpha T_\beta \in V := \langle 1, a, c \rangle$. For $c = \mathbf{1}_K$: $c^2 = c \in V$ (idempotent) and $a c = a_K c \in V$ (the indicator localises the product), so $\Pi_{cc} = \Pi_{ac} = 0$ —the c –flow lines are great circles and the surface is ruled by them. But $a^2 \notin V$ whenever a has ≥ 3 distinct values (Prop. 12.1), so

$\Pi_{aa} \neq 0$: the surface bends in the a -direction and is not totally geodesic. This is why the curvature is constant $\frac{1}{4}$ (the extrinsic curvature $\det \Pi = \langle \Pi_{aa}, \Pi_{cc} \rangle - |\Pi_{ac}|^2 = 0$) without the surface being a great subsphere.

Corollary 10.2 (the catalog surface). Since $\deg = 2 + 2 \cdot \mathbf{1}_{\{i=K\}}$ (only the quartic K , minimal polynomial $x^4 + 5x^2 - 5$, has degree 4) and $\log M$ is non-constant on the remaining seeds, the catalog's $(\log M, \deg)$ Gibbs surface satisfies Theorem 10.1 and has *exactly* constant curvature $\frac{1}{4}$. The multiplicity of φ, τ (equal $\log M$ and degree) is absorbed as a base-measure weight and does not affect the result. [FORCED]

The catalog surface, explicitly. At the anchor $\beta = -1$ the apex mass is $q_K = p_K = 0.2256$, giving latitude $\psi = \arcsin \sqrt{q_K} = 0.495$; the conditional distribution on the six non-apex seeds is $(0.0996, 0.1493, 0.2489, 0.0805, 0.0805, 0.3412)$, whose square roots are the point x on the equatorial S^5 . One checks $\cos^2 \psi = 1 - q_K = 0.7744$, so the catalog point sits on the suspension exactly as $\sqrt{p} = (\cos \psi x, \sin \psi)$. Sweeping β moves x along the $\log M$ -curve and ψ toward the apex K , tracing out the constant-curvature- $\frac{1}{4}$ sphere; the coordinate (ψ, ℓ) of Theorem 10.1, with ℓ the arc length of the $\log M$ -curve, realises it as a round sphere intrinsically (though not as a great subsphere of the ambient).

This inverts the naive expectation: the second statistic does not bend the geometry into a new varying-curvature object; the $(\log M, \deg)$ family sits inside the surprisal sphere as a constant-curvature ruled surface. The apex is essential—a generic two-statistic family (no indicator) has non-constant, sign-changing curvature; explicit controls give $4/21$ ($m = 4$) and $857/16928$ ($m = 5$), both $\neq \frac{1}{4}$. The general Brioschi curvature of an indicator family, its invariance under base measure (hence constancy), the ruled second fundamental form, and the suspension metric are all machine-verified (`suspension_theorem.py` and `field_surprisal_classification.py`).

Remark 10.3. Neither the trace form nor the $(\log M, \deg)$ surface is totally geodesic in the surprisal sphere—but for opposite reasons. Totally geodesic surfaces are the curvature- $\frac{1}{4}$ subspheres; the field's *flat* trace-form geometry misses this because its curvature is $0 \neq \frac{1}{4}$ (forcing extrinsic $k_1 k_2 = -\frac{1}{4}$, §12), while the $(\log M, \deg)$ surface has the right intrinsic curvature $\frac{1}{4}$ but the wrong embedding—it is ruled, not geodesic ($\Pi_{aa} \neq 0$). In fact (Prop. 12.1) *no* $(\log M, X)$ surface is totally geodesic.

11 The multi-statistic landscape: degree is geometrically distinguished

Theorem 10.1 singles out one property of the degree statistic—being an indicator of a *single* outcome. The catalog carries other forced invariants, and comparing their two-statistic surfaces shows how special that property is (Table 6).

The number of conjugates outside the unit circle is itself an indicator, $\#\{|\zeta| > 1\} = 2 - \mathbf{1}_{\{\varphi, \tau, \varphi^4\}}$ —but of a *three-element* set, not a single outcome. Its surface has non-constant, sign-changing curvature, so the single-outcome hypothesis of Theorem 10.1 is sharp: a set indicator makes the family a join of two non-trivial curves (one on the set's sub-sphere, one on its complement), which is not a suspension over a point, hence not a round sphere. The trace and house are generic (non-indicator) and likewise vary. Among the catalog's forced invariants, *degree is the unique one whose Gibbs surface with $\log M$ has constant curvature $\frac{1}{4}$* —a geometric distinction of the single quartic seed K , verified in `field_surprisal_tier2.py` (15/15).

statistic X	values	type	K of $(\log M, X)$ at three points
degree	$\{2, 4\}$	$\mathbf{1}_{\{K\}}$ (single outcome)	0.250, 0.250, 0.250
$\#\{ \zeta > 1\}$	$\{1, 2\}$	$\mathbf{1}_{\{\varphi, \tau, \varphi^4\}}$ (3-set)	0.020, 0.066, -0.006
trace	$\{-1, 0, 1, 7\}$	non-indicator	0.177, 0.056, 0.197
house $\max \zeta $	six values	non-indicator	0.211, 0.114, 0.227

Table 6: Gaussian curvature of the two-statistic Gibbs surface $(\log M, X)$ for four forced invariants, sampled at three natural-parameter points. Only the single-outcome indicator (degree) is the constant $\frac{1}{4}$. All [COMPUTED] (curvature routine validated on four metrics of known curvature).

12 The charge grading: a second constant-curvature- $\frac{1}{4}$ surface

The catalog carries a further forced invariant from the companion charge-measure coupling [10]: the conjugate-angle charge. An algebraic number is charge-admissible if every conjugate α satisfies $\alpha^n \in \mathbb{R}_{>0}$ for some $n \geq 1$; the least such n is its charge group $\mathbb{Z}/n\mathbb{Z}$, and the charge of a root at argument θ is $\chi_n = \text{round}(n\theta/2\pi) \bmod n$. Read off the catalog:

seed	$\sqrt{2}$	$\sqrt{3}$	$\sqrt{5}$	φ	τ	φ^4	K
charge group	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/1$	$\mathbb{Z}/4$

The seed φ^4 is the *unique* one with both conjugates real positive (roots φ^4, φ^{-4}), so its charge group is trivial $\mathbb{Z}/1$; K alone has complex conjugates $\pm i\beta$, giving the maximal $\mathbb{Z}/4$; the remaining five sit at $\mathbb{Z}/2$.

The paper’s central grading is charge *parity*, which sets the Mahler floor— φ for even n , 2 for odd n , the odd gap $(\varphi, 2)$ being the no-Salem closure [10, Thm. 5.1, 6.2]. For this catalog parity is $\mathbf{1}_{\{\varphi^4\}}$, a single-outcome indicator, so by Theorem 10.1 the surface $(\log M, \text{charge parity})$ again has constant curvature $\frac{1}{4}$ —verified at $K = 0.250, 0.250, 0.250$, and exactly, the Brioschi curvature of an arbitrary statistic with the φ^4 indicator being identically $\frac{1}{4}$ [FORCED]. The charge group *order* $\{1, 2, 4\}$ is three-valued and gives a non- $\frac{1}{4}$, varying surface $(0.339, 0.362, 0.342)$, consistent with §11.

We can now classify *all* such surfaces completely, and the answer corrects a natural over-expectation: constant curvature $\frac{1}{4}$ is strictly weaker than being totally geodesic (a great subsphere), and here only the former ever holds.

Proposition 12.1 (No surface is totally geodesic). *For every second statistic X , the $(\log M, X)$ surface is not totally geodesic in $S^6(2)$. Indeed a totally geodesic $V = \langle 1, \log M, X \rangle$ would be a unital subalgebra of $\mathbb{R}^7$, forcing the number of distinct value pairs $(\log M_i, X_i)$ down to $\dim V = 3$; but the cost takes six distinct values, so the pair count is always 6 or 7, never the 3 a great 2-sphere requires. [FORCED]*

Remark 12.2 (Correction). An earlier draft argued $\Pi_{aa} \neq 0$ from “ $\log^2 M \notin V$ ”; that inference is false as stated: for X affinely $\log^2 M$ one has $\log^2 M \in V$ and $\Pi_{aa} = 0$ (then $\Pi_{ac} \neq 0$ instead, since $\log^3 M \notin \langle 1, \log M, \log^2 M \rangle$; machine-checked in `t1_core.py`). The value-count argument above is the correct and complete proof; the proposition’s statement is unchanged.

Theorem 12.3 (Complete constant-curvature classification). *The $(\log M, X)$ surface has constant Gaussian curvature $\frac{1}{4}$ iff X is affinely the indicator $\mathbf{1}_S$ of a set S with $\log M$ constant on S or on its complement. There are exactly eight such surfaces (a surface depends only on the partition $\{S, S^c\}$): the seven single-seed indicators and the golden-pair merge $\mathbf{1}_{\{\varphi, \tau\}}$. Each is a ruled*

constant-curvature $-\frac{1}{4}$ surface (Theorem 10.1), none totally geodesic. No non-indicator X gives constant curvature $\frac{1}{4}$.

Proof. Sufficiency is Theorem 10.1 (idempotency gives $\Pi_{cc} = 0$; $\log M$ constant on S gives $\Pi_{ac} = 0$; hence $\det \Pi = 0$); the 16-subset census among all 127 indicators is exhaustive (field_surpraisal_classification.9/9). Necessity—no non-indicator X is constant $-\frac{1}{4}$ —is proved in full in §13 (Theorem 13.11); the earlier ≈ 5000 -sample evidence is retired and the tag upgrades to [FORCED].

Remark 12.4 (The eight surfaces and their forcing). Seven of the eight are single-seed, and each is *forced* by an arithmetic invariant:

seed s	$\sqrt{2}$	$\sqrt{3}$	$\sqrt{5}$	φ	τ	φ^4	K
singled out by	$M=2$	$M=3$	$M=5$	$\text{tr}=1$	$\text{tr}=-1$	$M=\varphi^4$	$M=\varphi^4-1$

The Mahler measure alone distinguishes five seeds; the only degeneracy is the golden pair $\{\varphi, \tau\}$ tied at $M = \varphi$, split by the trace ± 1 —the same sign the $\lambda = 2c$ construction uses to select the Perron keystone $R^2 = R + I$ (trace $+1$) over the Clifford conjugate gate $R^2 = I - R$ (trace -1) [11]. The eighth surface is precisely $\mathbf{1}_{\{\varphi, \tau\}}$, the single-outcome indicator of the *effective* catalog in which the Mahler-tied seeds φ, τ are merged. The two deepest of the seven are K and φ^4 , the charge-extreme seeds (charge group $\mathbb{Z}/4$ and $\mathbb{Z}/1$), additionally distinguished by degree, non-reality, and total positivity.

13 The classification completed: a finite branch calculus

Theorem 12.3 left one [OPEN]: ruling out every non-indicator statistic. We close it. Write $a = \log M$, $V = \langle 1, a, X \rangle$ (assumed 3-dimensional), and let Q_θ be the $L^2(p_\theta)$ -projection onto $V^\perp$ along the family $p \propto e^{\theta_1 a + \theta_2 X}$. The Gauss equation reduces constancy to one scalar: with $F = a^2$, $G = X^2$, $H = aX$,

$$q(\theta) = \langle Q_\theta F, Q_\theta G \rangle - \|Q_\theta H\|^2, \quad K \equiv \frac{1}{4} \iff q \equiv 0.$$

Lemma 13.1 (θ -rigidity). $\ker Q_\theta = V$ for every θ (the Gram matrix of $(1, a, X)$ is nonsingular under any positive weight). Hence $\Pi_{ac} \equiv 0 \iff aX \in V$ and $\Pi_{cc} \equiv 0 \iff X^2 \in V$: each ruled condition holds at one temperature iff at all. [FORCED]

Theorem 13.2 (The ruled system, solved). $\{aX \in V, X^2 \in V\} \iff X \cdot V \subseteq V \iff X$ is affinely $\mathbf{1}_S$ with a constant on S or S^c —exactly the eight families. Module route: $X \cdot V \subseteq V$ makes V a module over itself, hence a sum of level-set blocks of X ; $\dim V = 3$ against six distinct costs leaves two levels with a constant on one. Elimination route: $aX \in V$ makes X a Möbius function $\mu(a)$ off at most one pole level; the increment identity $\mu(s) - \mu(t) = (d_0 + d_1 d_2)(t - s) / \{(s - d_2)(t - d_2)\}$ and $X^2 \in V$ force μ constant off-pole, leaving pole-level indicators. [FORCED] (*t1_core.py*: the invariant module $N(X) = \{v \in V : vV \subseteq V\}$ is 2-dimensional exactly on the eight families, 1-dimensional on every control, including $X \simeq \log^2 M$ and the pair-pole family.)

Off the ruled locus, $q \equiv 0$ must hold with both forms nonzero. Two exact reductions make the question finite.

Proposition 13.3 (Numerator and the window identity). Put $C(f, g) = \sum_{i < j} w_i w_j (f_i - f_j)(g_i - g_j)$ on weights $w_i = e^{\theta_1 a_i + \theta_2 X_i}$, $D = C(a, a)C(X, X) - C(a, X)^2$, and $R(f, g) = DC(f, g) - C(f, \cdot) \text{adj } C \cdot C(\cdot, g)$. Then $Z^2 Dq = P(w) := R(F, G) - R(H, H)$, a degree-6 form whose 462

monomials all have support ≥ 4 . For a four-set s let ℓ_s be the signed 3×3 minors of the rows $(1, a, X)|_s$ and

$$q_4(s) = \ell_s(a^2) \ell_s(X^2) - \ell_s(aX)^2.$$

Then the coefficient of $w_h^3 w_j w_k w_l$ in P equals $q_4(s)$, cofactor exactly 1, for all 140 heavy placements; and on the four-point face itself $\langle Qf, Qg \rangle = \kappa \ell_s(f) \ell_s(g)$ with $\kappa > 0$, so $q = \kappa q_4(s)$ there. [FORCED] (`t1_engine.py`, `t1_reduction.py`.)

Theorem 13.4 (Master window identity: the coefficient law is Sylvester plus Cauchy–Binet). *As a polynomial identity in independent weights $w_1, \dots, w_m$ and arbitrary statistics (a, X) ,*

$$P(w) = Z^2 \sum_{|s|=4} q_4(s) w_s, \quad Z = \sum_i w_i, \quad w_s = \prod_{i \in s} w_i, \quad q_4(s) = \ell_s(a^2) \ell_s(X^2) - \ell_s(aX)^2,$$

where $\ell_s(f) = \det(\mathbf{1}, a, X, f)|_s$ realizes the signed-minor functional of Proposition 13.3. [FORCED]

Proof. Three classical steps. (i) $R(f, g)$ is the 3×3 determinant of the matrix $(C(u, v))_{u \in (f, a, X), v \in (g, a, X)}$: expanding along the first row reproduces $DC(f, g) - C(f, \cdot) \operatorname{adj} C(\cdot, g)$ term by term. (ii) With $\langle u, v \rangle_w = \sum_i w_i u_i v_i$ one has $C(u, v) = Z \langle u, v \rangle_w - \langle u, \mathbf{1} \rangle_w \langle \mathbf{1}, v \rangle_w$, the 2×2 minor of the mixed Gram matrix $B(f, g) = (\langle u, v \rangle_w)_{u \in (\mathbf{1}, f, a, X), v \in (\mathbf{1}, g, a, X)}$ bordered on its $(1, 1)$ entry $\langle \mathbf{1}, \mathbf{1} \rangle_w = Z$. Sylvester’s determinant identity with pivot Z [8] gives $\det(C(u, v)) = Z^{4-2} \det B(f, g)$ (reordering rows and columns simultaneously leaves determinants unchanged). (iii) Writing $B(f, g) = U W V^T$ with U, V the $4 \times m$ statistic matrices and $W = \operatorname{diag}(w)$, Cauchy–Binet [8] gives $\det B(f, g) = \sum_{|s|=4} w_s \ell_s(f) \ell_s(g)$. Hence $P = R(F, G) - R(H, H) = Z^2 \sum_s w_s [\ell_s(F) \ell_s(G) - \ell_s(H)^2]$. $\square$

Corollary 13.5 (Coefficient law and structure). (a) Every monomial of P has degree 6 and support ≥ 4 , with the exact law $[w^\kappa]P = \sum_{|s|=4, \mathbf{1}_s \leq \kappa} c(\kappa - \mathbf{1}_s) q_4(s)$, where $c(2e_i) = 1$ and $c(e_i + e_j) = 2$; the monomial census $140 + 210 + 105 + 7 = 462$ over the shapes $(3, 1, 1, 1), (2, 2, 1, 1), (2, 1, 1, 1, 1), (1^6)$ is forced. (b) The heavy coefficient $[w_h^3 w_j w_k w_l]P = q_4(s)$ with cofactor exactly 1: w_h^2 is the unique contribution of Z^2 . (c) Since the squarefree monomials $\{w_s\}$ are linearly independent, $q \equiv 0$ iff $q_4(s) = 0$ for all 35 windows—with no collision analysis. (d) $q_4(s) = -\prod_{t \in s} \det(\mathbf{1}, a, X)|_{s \setminus t}$, a product of the four triple minors, so the four-point trichotomy (Lemma 13.9) is immediate: $q_4(s) = 0$ iff some three of the four plane points are affinely dependent (collinear, or a coincident pair as at the golden merge). (e) $D = Z \sum_{|t|=3} w_t \Delta_t^2$ by the same two identities one order down, so $q = [\sum_s w_s q_4(s)] / [Z \sum_t w_t \Delta_t^2]$: the $\alpha = 0$ curvature obstruction is a ratio of consecutive Cauchy–Binet compound sums (the Hessian–metric picture of [6]), and restricting w to a four-point face recovers $q = \kappa q_4(s)$ with the explicit $\kappa = w_s / (Z \sum_{t \subset s} w_t \Delta_t^2) > 0$. [FORCED]

Remark 13.6 (Architecture; Open Problem 3 resolved). Theorem 13.4 replaces the machine discovery of Proposition 13.3 (140 exact divisions) by a two-line determinantal mechanism and resolves Open Problem (3) of the previous edition. The classification chain now runs rigidity $\rightarrow$ ruled system $\rightarrow$ Theorem 13.4 $\rightarrow$ trichotomy $\rightarrow$ plane lemma; the collision pencil and branch survival (Proposition 13.7, Lemma 13.8) are no longer load-bearing and are retained as an independent verification lane and as results on the collision structure in their own right. Verified end to end in `t1_windowproof.py` (44/44, exit 0): the chain symbolically at $m = 4, 5$; the factorization (d) symbolically for four generic points; exact rational instances at $m = 6, 7$; and, on the catalog ring, the full 462-coefficient dictionary equality of P with $Z^2 \sum_s q_4(s) w_s$, string-exact against the `t1_engine.py` artifact.

Proposition 13.7 (Collision pencil). $q \equiv 0$ iff the coefficients of P vanish after grouping monomials with equal frequency pairs $(\sum_i k_i a_i, \sum_i k_i X_i)$. Since $\log 2, \log 3, \log 5, \log \varphi$ are $\mathbb{Q}$ -linearly independent (multiplicative independence by norms in $\mathbb{Q}(\sqrt{5})$; no transcendence input), the cost collisions form a rank-2 lattice generated by the golden swap $e_\varphi - e_\tau$ and the Salem square $2e_K - e_{\sqrt{5}} - e_{\varphi^4}$, reflecting $(\varphi^4 - 1)^2 = 5\varphi^4$. Every X -collision condition lies on the pencil $j\Delta_1 + k\Delta_2 = 0$ with $\Delta_1 = X_\varphi - X_\tau$, $\Delta_2 = 2X_K - X_{\sqrt{5}} - X_{\varphi^4}$; the realizable primitive relations are only Δ_1 , Δ_2 , $\Delta_1 \pm \Delta_2$, $2\Delta_1 \pm \Delta_2$, and two independent relations force $\Delta_1 = \Delta_2 = 0$. [FORCED]

Lemma 13.8 (Branch survival). In every branch of the pencil—generic, each of the five nontrivial single relations, and both sub-branches of the merged six-point recursion for $\Delta_1 = 0$ (whose own collision lattice has rank 1, again the Salem square; the base weight 2 at the merged point is invisible to vanishing since the collision support misses it)—every window s retains at least one heavy placement whose group is a singleton. Hence $q \equiv 0$ forces $q_4(s) = 0$ for all windows in every branch. [FORCED] (*t1.branches.py*: zero windows lost across all eight branch runs; all 35 windows vanish identically on the eight families in symbolic (α, β, γ) , and a random control violates one.)

Lemma 13.9 (Four-point trichotomy). $q_4(s) = 0$ iff some three of the four plane points $T_i = (a_i, X_i)$, $i \in s$, are collinear, or $\{\varphi, \tau\} \subset s$ with $T_\varphi = T_\tau$. Collinear triples are automatically non-vertical, the cost having a single tie.

Proof. q_4 is the determinant of the face's residual scalar form; it vanishes iff $\text{rank}(1, a, X)|_s \leq 2$ (all four collinear) or a null direction $v = \alpha a + \beta X$ has $vV|_s \subseteq V|_s$. Diagonalizing v : four distinct v -levels are impossible (**1** has full support); three levels, or a 2+2 split, force the tied pair constant in both coordinates, i.e. $T_\varphi = T_\tau$; a 1+3 split puts X affine in a on the triple. The ray $\beta = 0$ reduces to $V|_s = \langle 1, a, a^2 \rangle$ and the minimal polynomial of $a|_s$, landing in the same two alternatives. $\square$

Lemma 13.10 (Plane lemma). If $n \geq 5$ plane points are such that every 4-subset contains a collinear triple, then all but at most one lie on a single line.

Proof. Let L be a line through the most points, $|L| \geq 3$. Suppose $p \neq q$ lie off L . For any $x, y \in L$, the 4-set $\{p, q, x, y\}$ contains a collinear triple; $\{p, x, y\}$ or $\{q, x, y\}$ would put p or q on L , so $\text{line}(p, q)$ contains x or y —for every pair $x, y \in L$. Taking three points of L pairwise, $\text{line}(p, q)$ would contain two of them and meet L twice: contradiction. $\square$

Theorem 13.11 (Complete classification, closed). $q \equiv 0$ iff X is affinely $\mathbf{1}_S$ with S one of the seven seeds or the golden pair. Theorem 12.3 therefore holds as stated with tag [FORCED]: the eight surfaces are the only constant-curvature- $\frac{1}{4}$ two-statistic surfaces.

Proof. If $\Delta_1 \neq 0$ the coincidence alternative of Lemma 13.9 cannot fire, so by Lemmas 13.8–13.10 at least six of the seven T_i lie on one non-vertical line ℓ , i.e. $X = \alpha + \beta a + \gamma \mathbf{1}_S$ with S the exceptional point. Both golden points on ℓ would give $X_\varphi = X_\tau$ (equal abscissae on a non-vertical line), contradicting $\Delta_1 \neq 0$; so the exception is φ or τ , automatically an a -level. If $\Delta_1 = 0$ the family is the merged six-point catalog, whose six costs are distinct; the same window-trichotomy-plane chain on its 15 windows puts five of six points on a line, giving the merged singletons—pulling back to the golden merge, or a singleton with $X_\varphi = X_\tau$. All cases land in the eight families; the converse is Theorem 10.1. $\square$

Remark 13.12 (Below the ceiling). For any subset S , idempotency gives $\Pi_{cc} = 0$, hence $\det \Pi = -|\Pi_{ac}|^2 \leq 0$ and $K \leq \frac{1}{4}$ pointwise: every indicator surface sits at or below the $\frac{1}{4}$ ceiling, the eight families being exactly the equality case—consistent with the 3-set indicator row of Table 6.

14 Product geometry: independent catalogs do not curve each other

Two catalogs Ω_1, Ω_2 with their own temperatures give the product family $p_{ij}(\beta_1, \beta_2) \propto M_i^{(1)-\beta_1} M_j^{(2)-\beta_2} = p_i^{(1)}(\beta_1) p_j^{(2)}(\beta_2)$. Its log-partition splits, $A(\beta_1, \beta_2) = A_1(\beta_1) + A_2(\beta_2)$, so the Fisher matrix is block-diagonal:

$$\nabla^2 A = \begin{pmatrix} \text{Var}_{\beta_1}(\log M^{(1)}) & 0 \\ 0 & \text{Var}_{\beta_2}(\log M^{(2)}) \end{pmatrix}, \quad \frac{\partial^2 A}{\partial \beta_1 \partial \beta_2} = 0;$$

the two fields' description costs are statistically independent.

Proposition 14.1 (No cross-curvature). *The joint categorical model of two independent catalogs is the Riemannian product $S^{m_1-1}(2) \times S^{m_2-1}(2)$: its sectional curvature is $\frac{1}{4}$ within each spherical factor and 0 on every mixed 2-plane; equivalently, all Christoffel symbols mixing the two blocks vanish.* [FORCED]

Proof. The product metric is block-diagonal with each block a function of its own coordinates only; hence every Christoffel symbol with indices spanning both blocks vanishes, so does every mixed Riemann component, and the sectional curvature of a plane spanning one direction from each factor is 0 (verified in `field_surprisal_tier2.py`). Each factor keeps its intrinsic curvature $\frac{1}{4}$ (Lemma 4.3). $\square$

To make two fields interact geometrically—nonzero cross-curvature—one must replace the product by a warped metric $g_1 \oplus f(\cdot)^2 g_2$ (or a nontrivial fibration), introducing a coupling function f . No invariance selects f : by the same Čencov argument that leaves the temperature β (and the companion paper's conformal scale c) undetermined, the coupling is a *declared* datum [DECLARED], of exactly that epistemic status. The product of the two Gibbs *curves* (one temperature each) is, by contrast, flat—a product of one-dimensional manifolds—a degenerate special case carrying no field interaction at all.

15 The temperature is a coordinate, not a parameter

The ledger has carried one [DECLARED] since §3: the Gibbs form $p \propto M^{-\beta}$ and the choice of β . Product functoriality dissolves the first half.

Theorem 15.1 (Forced form). *Let $p_i \propto f(M_i)$ depend on the seeds through the Mahler measure alone, with $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ Lebesgue measurable, and require product functoriality: on every product catalog the joint weight matrix $[f(M_i M'_j)]$ factorizes as a product measure. Then $f(x) = c x^{-\beta}$: the Mahler–Gibbs family is forced, and β is the canonical affine coordinate of its e -geodesic.* [FORCED] (modulo the stated regularity)

Proof. Factorization is the rank-1 condition on $[f(M_i M'_j)]$, i.e. $f(xy) = f(x)f(y)/f(1)$ on the value grid; ranging over finite mass sets this is Cauchy's multiplicative equation on $\mathbb{R}_{>0}$, whose measurable solutions are the power laws [7]. The regularity is genuinely necessary: on the bare multiplicative lattice generated by $\{2, 3, 5, \varphi\}$, independent per-generator exponents are multiplicative without being any single power (`t2_temperature.py` exhibits one exactly), so “a single β ” is precisely the content of measurability. $\square$

What remains declared is only the selection of an operating *point* on a forced curve, and the curve's marked points are each forced by a separate principle:

β	forced by	$I(\beta)$	$s(\beta)$
-1	anchor $Z(-1) = 17 + 4\sqrt{5}$	0.26260	-0.5539
$\beta^* \approx -0.07677$	max Fisher ($I' = -\kappa_3 = 0$)	0.32886	-0.0440
0	maximum entropy	0.32832	0
1	MDL	0.24304	+0.5430
$\sqrt{5}$	exchange rate $\lambda = 2c$	0.10405	+1.0461
$\beta_C \approx 2.5455$	heat-capacity peak ($\beta > 0$)	—	+1.1401

with $Z(+1) = \sum M^{-1} = \frac{91}{30} - \frac{\sqrt{5}}{5}$ exact. [FORCED] for the anchors and the identity $I'(\beta) = -\kappa_3(\beta)$; [COMPUTED] for the numeric column.

Proposition 15.2 (Finite total information length). *$I(\beta) \rightarrow 0$ exponentially in both directions (spectral-gap concentration), so the Gibbs curve has finite total Fisher–Rao length*

$$L_{\text{tot}} = \int_{-\infty}^{\infty} \sqrt{I(\beta)} d\beta \approx 5.64622,$$

an intrinsic invariant of the catalog. Its endpoints are forced: $\beta \rightarrow +\infty$ concentrates on the cost minimizers—the golden merge $\{\varphi, \tau\}$, uniform—and $\beta \rightarrow -\infty$ on the cost maximizer, the φ^4 vertex. The forced e-geodesic runs from the eighth surface's apex to the charge-trivial seed, and arc length s is its canonical parametrization; the segment $[-1, \sqrt{5}]$ carrying all six marked points uses only 28.3% of the total length. [FORCED] for the limits; [COMPUTED] for the values (`t2_temperature.py`).

Theorem 15.3 (Selection dichotomy). *The metric selects no operating point: a finite-length one-dimensional Riemannian curve has isometry group $\{\text{id}, \text{flip}\}$, and the flip swaps the endpoints, which are distinguishable (the $\beta \rightarrow +\infty$ limit is uniform on the golden pair, the $\beta \rightarrow -\infty$ limit is the φ^4 vertex); the labeled-endpoint isometry group is therefore trivial, and no interior point is metrically canonical. [FORCED] Meanwhile the invariant principles on offer provably disagree: with $s(0) = 0$, the arc-length midpoint sits at $s(\beta_{\text{mid}}) \approx -0.3401$, the max-Fisher point at $s(\beta^*) \approx -0.0440$, maximum entropy at 0, and the heat-capacity peak at $s(\beta_C) \approx +1.1401$: pairwise distinct with margins far above numeric error. [COMPUTED] Selecting an operating temperature is therefore irreducibly an election among inequivalent invariant principles; the [DECLARED] on the selection is not a gap awaiting closure but the honest form of the answer (`t6_selection.py`, 18/18: exact $\mathbb{Q}(\sqrt{5})$ anchors, symbolic $I' = -\kappa_3$, dps-30/50 stability).*

16 Higher statistics: the iterated join and the geodesic threshold

Theorem 16.1 (Iterated join). *Let $s_1, \dots, s_{k-1}$ be distinct outcomes and a a non-constant on the complement. The k -parameter family $p \propto \nu \exp(\theta_0 a + \sum_j \theta_j \mathbf{1}_{s_j})$ maps under $\sqrt{p}$ onto the spherical join of the totally geodesic corner S^{k-2} spanned by the apexes with the base curve of the a -family on the complement; in join coordinates the metric is*

$$d\Psi^2 + \sin^2\Psi g_{S^{k-2}} + \cos^2\Psi d\ell^2,$$

of constant sectional curvature 1. The family is therefore, isometrically, an open subset of a round $\frac{1}{4}$ -sphere of dimension k : the suspension theorem iterates in every dimension. [FORCED] (`t3_suspension.py`: $R_{ijkl} = g_{ik}g_{jl} - g_{il}g_{jk}$ exactly for $k = 3, 4$; the pullback of the round metric under the join map is computed symbolically.)

Corollary 16.2 (The catalog three-fold). $(\log M, \deg, \text{charge parity}) = (\text{cost}, \mathbf{1}_K, \mathbf{1}_{\varphi^4})$ is a three-dimensional round $\frac{1}{4}$ -sphere: cost, degree structure, and total positivity together still curve exactly like the simplex. Sampled sectional curvatures agree with $\frac{1}{4}$ to 10^{-18} over random natural parameters and 2-planes (routine validated on four metrics of known curvature); the family is not totally geodesic ($\log^2 M \notin V$). [FORCED]/[COMPUTED]

Theorem 16.3 (Geodesic threshold). *A family $V \ni \log M$ is totally geodesic in $S^6(2)$ iff V is a unital subalgebra of $\mathbb{R}^7$. Any such V contains $\mathbb{R}[\log M]$, which is six-dimensional and equals the algebra of functions constant on the golden pair. Hence no totally geodesic family with $k \leq 4$ statistics exists; the first appears at exactly $k = 5$, uniquely $V = \mathbb{R}[\log M]$ —the effective-catalog simplex, a great S^5 realizing the golden merge as forced geometry—and $k = 6$ is the full simplex.* [FORCED]

17 The k -window calculus and the $k = 3$ census

The master identity lifts to every k by the same three steps, and the lift carries a complete classification of a first $k = 3$ landscape.

Theorem 17.1 (Generalized window identity). *Let $T_1, \dots, T_k$ be statistics with $\dim V = k + 1$, $V = \text{span}(\mathbf{1}, T)$, and for $|s| = k + 2$ put $\ell_s(f) = \det[(\mathbf{1}, T_1, \dots, T_k, f)|_s]$. With $C(u, v) = Z\langle u, v \rangle_w - \langle u, \mathbf{1} \rangle_w \langle \mathbf{1}, v \rangle_w$, Sylvester's identity with pivot Z and Cauchy–Binet give, exactly as at $k = 2$,*

$$\det[C(u_i, v_j)]_{(k+1) \times (k+1)} = Z^k \sum_{|s|=k+2} w_s \ell_s(f) \ell_s(g), \quad D_k \langle Qf, Qg \rangle_p = Z^{k-2} \sum_{|s|=k+2} w_s \ell_s(f) \ell_s(g),$$

over rows $u = (f, T)$, columns $v = (g, T)$, with $D_k = \det[C(T_a, T_b)] = Z^{k-1} \det \text{Gram}_w(\mathbf{1}, T)$ and Q the $L^2(p)$ -projection off V . [FORCED] (**t4.kwindows.py**: fully symbolic at $k = 3$, $m = 5$; exact rational instances $m = 6, 7$; both Sylvester scalings.)

Proposition 17.2 (Windowed Gauss obstructions and the flat criterion). *The curvature- $\frac{1}{4}$ obstructions $G^{abcd} = \langle Q(T_a T_c), Q(T_b T_d) \rangle_p - \langle Q(T_a T_d), Q(T_b T_c) \rangle_p$ satisfy*

$$D_k G^{abcd} = Z^{k-2} \sum_{|s|=k+2} w_s [\ell_s(T_a T_c) \ell_s(T_b T_d) - \ell_s(T_a T_d) \ell_s(T_b T_c)],$$

the right side running over the 2×2 minors of the window matrix $L(s) = [\ell_s(T_a T_b)]$. If $\dim V = k + 1$ then $h \in V \iff \ell_s(h) = 0$ for every window (flat lemma). Hence: if every window matrix has rank ≤ 1 , all Gauss obstructions vanish identically and the family has constant sectional curvature $\frac{1}{4}$. [FORCED]

Corollary 17.3. The catalog three-fold $(\log M, \mathbf{1}_K, \mathbf{1}_{\varphi^4})$ has all 21 window matrices of rank ≤ 1 identically over $\mathbb{Q}(\log 2, \log 3, \log 5, \log \varphi)$, and the four-fold $(\log M, \mathbf{1}_K, \mathbf{1}_{\varphi^4}, \mathbf{1}_{\sqrt{5}})$ has all 7: exact linear-algebra reconfirmations of Corollary 16.2 and of Theorem 16.1 at $k = 4$, with no curvature routine involved. [FORCED]

Theorem 17.4 (The $k = 3$ double-indicator census). *Consider triples $(\log M, \mathbf{1}_S, \mathbf{1}_T)$ over pairs of nonempty proper subsets with $\dim V = 4$ (7812 of the 7875 unordered pairs). Windowwise flatness reduces, by rank ≤ 1 on the symmetric window matrix and the flat lemma, to three exact memberships $\{a\mathbf{1}_S, a\mathbf{1}_T, \mathbf{1}_{S \cap T}\} \subset V$ over $\mathbb{Q}(\log 2, \log 3, \log 5, \log \varphi)$. Exactly 312 pairs pass, and they collapse onto exactly **26** distinct spans*

$$V = \langle \mathbf{1}, \log M, \mathbf{1}_{S'}, \mathbf{1}_{T'} \rangle, \quad S', T' \text{ disjoint subsets each inside a single cost level}$$

$\binom{8}{2} - 2$: eight within-level subsets, the two golden overlaps being redundant presentations of the all-golden class, where $\mathbf{1}_\varphi + \mathbf{1}_\tau = \mathbf{1}_{\varphi\tau}$ collapses). Each of the 26 families is constant-curvature $\frac{1}{4}$ by Proposition 17.2, and the eight $k = 2$ surfaces embed as the pairs with one within-level generator. [FORCED] (Two lanes: `t4_kwindows.py` decides membership by symbolic rank over the coefficient field; `t4b_census_fast.py` re-derives the block-collapsed linear system and decides by exact $\mathbb{Q}^5$ -vector annihilator tests with quadratic cross-ratio consistency; sampled cross-lane agreement on 24 pairs $\times$ 3 targets.)

Remark 17.5. What does *not* lift verbatim is the $k = 2$ chain “windows $\Rightarrow$ trichotomy $\Rightarrow$ plane lemma”: the multiset analog of the plane lemma is false at $k \geq 3$ (a doubled support point defeats the general-position projection). Both halves of the $k \geq 3$ program nonetheless close, by different routes: necessity in §19 (squarefree collisions and twin symmetry), and the landscape in §21, where the moment trichotomy makes the doubled point harmless—a 2-circuit never obstructs—and the plane lemma is bypassed rather than repaired (Remark 21.5). The census above is the double-indicator stratum of the full landscape (Theorem 21.3).

Open Problem 17.6. *Resolved in this edition, in both parts.* Necessity: Theorem 19.2—constant $\frac{1}{4}$ is exactly windowwise rank ≤ 1 . Landscape: §21—constant $\frac{1}{4}$ is exactly the partitioned-affine condition $V = V(\pi)$, with catalog counts 8/56/95/31/1 for $k = 2, \dots, 6$; the conjecture of the previous edition, that within-level indicator joins exhaust, is *refuted* by the 30 split-affine classes $\langle 1, a, \mathbf{1}_B, a\mathbf{1}_B \rangle$ (Corollary 21.4).

18 Catalog invariance: the census theorem and the branch-rank calculus

The eight-surface census is not an artifact of the seed selection: its shape is a theorem about cost coincidences, and the coincidences are localized exactly.

Theorem 18.1 (Generic census). *If the costs $a_1, \dots, a_m$ are $\mathbb{Q}$ -linearly independent, the collision lattice $\{d \in \mathbb{Z}^m : \sum d_i a_i = 0, \sum d_i = 0\}$ is zero; by the master identity the vanishing $q \equiv 0$ then forces every monomial coefficient of P individually, hence every window $q_4(s) = 0$, hence (trichotomy and plane lemma, both catalog-free) exactly the m single-seed surfaces. No branch analysis occurs.* [FORCED]

Theorem 18.2 (Census count and catalog invariance). *For any catalog, every within-level indicator $\mathbf{1}_S$ (S inside one cost level) is a nondegenerate constant- $\frac{1}{4}$ surface: in the factorization $q_4 = -\prod \Delta$ each window carries a vanishing factor (a coincident pair if $|s \cap S| \geq 2$, three collinear points on $X = 0$ otherwise). When the necessity analysis closes, the census is exactly*

$$\#\{\text{constant-}\frac{1}{4}\text{ surfaces}\} = \sum_{\text{levels } \ell} (2^{m_\ell} - 1),$$

so the “eighth surface” is precisely the golden coincidence’s $2^2 - 1 = 3$. Verified end to end for four catalogs: the full catalog (collision lattice of rank 2: golden swap Δ_1 , Salem square Δ_2 ; census 8), drop- K (rank 1, golden; 7), drop- τ (rank 1, Salem; 6), and add- $\sqrt{7}$ (rank 2 embedded; 9). [FORCED] (`t5_catalog_census.py`, 72/72.)

Theorem 18.3 (Branch-rank calculus). *Necessity closes for an arbitrary catalog by a rank criterion. The realizable primitive relation directions D are the primitive lattice vectors d with k and $k+d$ both monomials of P (for the full catalog, $D = \{\Delta_1, \Delta_2, \Delta_1 \pm \Delta_2, 2\Delta_1 \pm \Delta_2\}$); the branches are*

the flats of the arrangement they span. Per branch, group the monomials of $P = Z^2 \sum_s q_4(s) w_s$ by (cost class, X -class mod the active relations): $q \equiv 0$ forces each grouped $\mathbb{Z}$ -combination of window symbols to vanish, and

$$\text{rank}_{\mathbb{Q}}(M_{\text{branch}}) = \#\{\text{windows}\} \implies \text{every } q_4(s) \text{ forced to } 0.$$

A coincidence direction $\pm(e_i - e_j)$ with $a_i = a_j$ merges the two seeds as plane points: windows containing both are automatically zero (coincident pair), the remaining symbols identify pairwise, and the rank test runs against the $\binom{m-1}{4}$ surviving classes. This upgrades the $k = 2$ singleton-survival lemma of §13 to a complete criterion (a unit heavy row per window is the special case), mechanically reproduces the eight-branch tree of the full catalog, and closes every branch of all four catalogs, with a one-sided mod- p rank certificate and an exact $\mathbb{Q}$ fallback. [FORCED]

19 k -necessity: constant $\frac{1}{4}$ is exactly windowwise flatness

Remark 17.5 left necessity at $k \geq 3$ open: the multiset plane lemma fails, so the $k = 2$ route does not lift. It turns out necessity never needed that route. The observation is that the exponential identity behind the Gauss obstructions lives at the *window* level, where every monomial is squarefree—and the catalog's collision lattice has only one squarefree direction.

Theorem 19.1 (Window collisions are golden). *Let $T = (\log M, T_2, \dots, T_k)$ be any statistics on the catalog, $2 \leq k \leq 5$. Two distinct $(k+2)$ -windows $s \neq s'$ have equal frequency vectors $\sum_{i \in s} T_i = \sum_{i \in s'} T_i$ iff $s' = s \triangle \{\varphi, \tau\}$ and every statistic ties on the golden pair, $T_{j,\varphi} = T_{j,\tau}$ for all j . [FORCED]*

Proof. The difference $\delta = \mathbf{1}_s - \mathbf{1}_{s'}$ has entries in $\{-1, 0, 1\}$ and zero sum, and the cost coordinate forces δ into the zero-sum collision lattice $\langle d_1, d_2 \rangle$ of Proposition 13.7 (d_1 the golden swap, d_2 the Salem square). The general element $j d_1 + k d_2 = (0, 0, -k, j, -j, -k, 2k)$ carries the entry $2k$ at K , so a cube point forces $k = 0$ and $|j| \leq 1$: $\delta = \pm d_1$, i.e. $s' = s \triangle \{\varphi, \tau\}$. The swap then preserves the remaining frequency coordinates iff every statistic ties on the pair. $\square$

Theorem 19.2 (k -necessity; the equivalence). *On the catalog, for $2 \leq k \leq 5$ and statistics $T = (\log M, T_2, \dots, T_k)$ with $\dim V = k + 1$: the family has constant sectional curvature $\frac{1}{4}$ iff every window matrix $L(s) = [\ell_s(T_a T_b)]$ has rank ≤ 1 . [FORCED] (`t7_knecessity.py`, 28/28.)*

Proof. Sufficiency is Proposition 17.2. For necessity, constant $\frac{1}{4}$ means the Gauss obstructions G^{abcd} of Proposition 17.2 vanish identically: they form an algebraic curvature tensor, so vanishing of all sectional values forces the full tensor by polarization. Since $D_k > 0$ and $Z > 0$ on the family, Proposition 17.2 turns $G^{abcd} \equiv 0$ into the exponential identity

$$\sum_{|s|=k+2} w_s(\theta) M^{abcd}(s) \equiv 0, \quad M^{abcd}(s) = \ell_s(T_a T_c) \ell_s(T_b T_d) - \ell_s(T_a T_d) \ell_s(T_b T_c),$$

whose coefficients range, over all quadruples, over exactly the 2×2 minors of the symmetric matrix $L(s)$ (rows $\{a, b\}$, columns $\{c, d\}$). If some statistic separates the golden pair, Theorem 19.1 gives pairwise distinct frequencies, so every $M^{abcd}(s) = 0$ individually. If every statistic ties, the pair are *twins*: for a swap pair the bordered matrices $(\mathbf{1}, T, T_a T_b)|_s$ and $(\mathbf{1}, T, T_a T_b)|_{s'}$ agree row for row (the twins occupy the same sorted slot, the indices being adjacent), so $\ell_{s \triangle \{\varphi, \tau\}} = \ell_s$ exactly and the grouped coefficient is $2M^{abcd}(s)$, forcing each; and a window containing both twins carries two identical rows, so $L(s) = 0$ outright. In every case all 2×2 minors of every $L(s)$ vanish, i.e. $\text{rank } L(s) \leq 1$. $\square$

Corollary 19.3 (Open Problem 17.6(i) closed; the census upgraded). Constant $\frac{1}{4}$ and windowwise flatness coincide unconditionally, so Theorem 17.4’s 26 classes are exactly the constant- $\frac{1}{4}$ double-indicator triples—the complete such landscape—and Corollary 17.3’s three- and four-folds need no curvature routine. The analytic classification problem is reduced, at every k , to finite linear algebra. [FORCED]

Remark 19.4 (Squarefreeness; the branch tree retires). Windows are squarefree monomials; the Salem square is a multiplicity-two relation and cannot act at window level—the same mechanism that made the heavy cofactor exactly 1 in Corollary 13.5(b). At $k = 2$ this collapses the eight-branch analysis to the golden dichotomy above: the branches lived on the *expanded* monomials of P , where the prefactor Z^2 reintroduces multiplicities; on the family one divides Z^2 out first. The branch machinery (§13, Theorem 18.3) is retained as an independent verification lane and as the general-catalog tool.

Theorem 19.5 (General catalogs). *Call a catalog transposition-tame if its zero-sum cost kernel meets $\{-1, 0, 1\}^m$ only in signed sums of disjoint equal-cost transpositions. For such catalogs the equivalence of Theorem 19.2 holds at every k , by the same twin argument classwise. All four studied catalogs qualify: full, drop- K , and add- $\sqrt{7}$ (cube points $\pm d_1$ only), and drop- τ (no cube points at all—no window collisions whatsoever). A catalog with a genuine $\{-1, 0, 1\}$ trade (e.g. costs with $a_1 + a_4 = a_2 + a_3$, all distinct) falls instead to the window-level branch-rank test—group windows by frequency class and require rank = #windows—the k -window analog of Theorem 18.3, strictly smaller than its monomial-level version.* [FORCED]

What remained of Open Problem 17.6—its part (ii), the landscape beyond double indicators—is closed in §21: the answer is the partitioned-affine classification, and the conjecture recorded below it fails.

20 The compositum couples: forced interaction between fields

Open Problem (4) asked whether the compositum’s own arithmetic forces a nonzero cross-structure between two fields, or whether they remain geometrically independent even there. It resolves on the first horn, with an exact witness.

For catalog seeds p, q let $p \otimes q$ be the *tensor polynomial*, whose roots are all conjugate products $\alpha_i \beta_j$ —no root is chosen, so the construction is gauge-free—and let $c(p, q) = \log M(p \otimes q)$ be the compositum cost. Every conjugate log-magnitude of the catalog is a $\mathbb{Q}$ -vector over $(\log 2, \log 3, \log 5, \log \varphi)$: for the quartic, $|K_{\text{out}}|^2 = \sqrt{5} \varphi^2$ and $|K_{\text{in}}|^2 = \sqrt{5} \varphi^{-2}$. Hence each $c(p, q)$ is an exact $\mathbb{Q}$ -vector; products on the unit circle (zero vector, e.g. $\varphi\psi = -1$) are excluded symbolically, and every remaining sign is certified by rigorous interval arithmetic.

Theorem 20.1 (The compositum couples). *The compositum cost is not additively separable: the interaction contrasts*

$$c(\sqrt{2}, \sqrt{3}) + c(\varphi, \varphi) - c(\sqrt{2}, \varphi) - c(\varphi, \sqrt{3}) = \log 2, \quad c(\varphi, \varphi) + c(\varphi^4, \varphi^4) - 2c(\varphi, \varphi^4) = -6 \log \varphi,$$

both exactly, as nonzero coefficient vectors. Consequently the compositum’s Mahler-Gibbs family $p \propto M(p \otimes q)^{-\beta}$ factors through the marginal families for no $\beta \neq 0$; its Fisher geometry carries nonzero cross-structure, $\text{Cov}_\beta(\text{cost}_1, \text{cost}_2) \neq 0$ at the marked temperatures [COMPUTED]; and the coupling left [DECLARED] by the product construction is, in the compositum reading, forced. [FORCED] (*t8.compositum.py*, 13/13.)

Proposition 20.2 (Structure of the coupling). (i) The rational sector $\{\sqrt{2}, \sqrt{3}, \sqrt{5}\}$ is exactly separable: $c(\sqrt{D}, \sqrt{D'}) = 2\log D + 2\log D'$, and all its interaction contrasts vanish. (ii) The golden tie persists: $c(\varphi, \cdot) = c(\tau, \cdot)$ (the conjugate magnitude multisets agree), so the compositum inherits the golden merge. (iii) The double-centered interaction tensor Δ is nonzero of exact rank 5 over $\mathbb{Q}(\log 2, \log 3, \log 5, \log \varphi)$: five independent coupling modes. The coupling is a golden/quartic phenomenon; the fields do not decouple anywhere off the rational sector. [FORCED]

Proposition 20.3 (Charge tensor law, instance-verified). With seed charge groups ($\mathbb{Z}/2$ at $\sqrt{2}, \sqrt{3}, \sqrt{5}, \varphi, \tau$; $\mathbb{Z}/1$ at φ^4 ; $\mathbb{Z}/4$ at K), every tensor conjugate is real or purely imaginary, and on all 28 unordered pairs the tensor polynomial's charge group is exactly $\mathbb{Z}/\text{lcm}(n_1, n_2)$ —e.g. $\sqrt{2} \otimes K \rightarrow \mathbb{Z}/4$, $\varphi^4 \otimes \varphi^4 \rightarrow \mathbb{Z}/1$ —consistent with the tensor charge law of the charge-measure companion. [FORCED] (finite instance check)

Remark 20.4 (Relation to the product geometry). Proposition 14.1 stands: the declared product of two catalogs is a Riemannian product with zero cross-curvature. The theorem says the compositum does not instantiate that construction: its forced costs are not the product costs, and the discrepancy is the exact tensor Δ . What the product section left as a declared warp is here a forced nonzero datum. The two questions this leaves—the coupled family's geometry, and whether Δ is charge-determined—are settled in §22.

21 The landscape completed: the partitioned-affine classification

With Theorem 19.2, constant $\frac{1}{4}$ is windowwise flatness; what Open Problem 17.6 still asked for was the classification of the windowwise-flat spans $V \ni 1, a$ with $\dim V = k + 1$. This section completes it, at every k , and the conjecture recorded there turns out to be *false*. The right chart is the point configuration: write $P_i = (T_1, \dots, T_k)(i) \in \mathbb{R}^k$ for the statistic values at seed i (with $T_1 = a$). For a window s whose points affinely span $\mathbb{R}^k$, the functional ℓ_s is exactly the unique affine dependency λ of the $k + 2$ points—supported on the window's unique circuit—and

$$L(s) = [\ell_s(T_a T_b)]_{a,b} = \sum_{i \in s} \lambda_i P_i P_i^\top,$$

whose rank is invariant under affine changes of the statistic basis (the cross terms die on $\sum \lambda_i = 0$, $\sum \lambda_i P_i = 0$). Non-spanning windows carry $L(s) = 0$ and no condition, so the window criterion is a statement about circuits.

Theorem 21.1 (Moment trichotomy). Let s be a spanning window with unique circuit C and dependency λ . If $|C| = 2$ (a coincident pair), $L(s) = 0$. If $|C| = 3$ (a collinear triple with parameters t_1, t_2, t_3), $L(s)$ has rank exactly 1, with moment $\sum_i \lambda_i t_i^2 = -(t_1 - t_2)(t_2 - t_3)(t_3 - t_1) \neq 0$. If $|C| \geq 4$, $\text{rank } L(s) \geq 2$. Consequently a k -family is constant- $\frac{1}{4}$ if and only if every circuit of its configuration has size ≤ 3 . [FORCED]

Proof. Only $|C| \geq 4$ needs an argument. A circuit's points are in general position in their $(d = |C| - 2)$ -flat; affinely normalize $d + 1$ of them to $0, e_1, \dots, e_d$, so the last is $(x_1, \dots, x_d)$ and $\lambda = (\sum_j x_j - 1, -x_1, \dots, -x_d, 1)$, all entries nonzero on a circuit. Then $Q = \sum \lambda_i P_i P_i^\top$ has $Q_{jj} = x_j^2 - x_j$ and $Q_{jl} = x_j x_l$. For $d \geq 3$ the mixed 2×2 minor on rows $\{j, l\}$, columns $\{l, r\}$ equals $x_j x_l x_r$; for $d = 2$, $\det Q = xy(1 - x - y)$. Both are products of circuit weights, hence nonzero. (t9, blocks 1–2: symbolic at $d = 2, 3, 4$, plus exact instances.) $\square$

Theorem 21.2 (Structure: flat = partitioned–affine). *Suppose the configuration is windowwise flat. Then its maximal collinear location–sets with ≥ 3 locations (rich lines) are pairwise disjoint and pairwise skew; every collinear triple of locations lies inside a single rich line; and, writing π for the partition of the seeds into rich–line blocks and free locations (coincident clusters and singletons),*

$$V = V(\pi) := \{f : f \text{ is affine in } a \text{ on every block of } \pi\}.$$

Conversely, every $V(\pi)$ with $\dim V(\pi) = k + 1$ is windowwise flat. [FORCED]

Proof. Two rich lines sharing a location or lying in a common plane yield four coplanar locations, two per line, with no collinear triple—a 4–circuit—contradicting Theorem 21.1; so rich lines are disjoint and skew. A collinear triple across structures would make its own line rich and meeting another. On each rich line the cost is affine and nonconstant (three locations in one level would need level multiplicity ≥ 3), hence an injective parameter, and every statistic is affine along the line; so $V \subseteq V(\pi)$. If $\dim V < \dim V(\pi) = 2r + f$, the affine hull is deficient and a transversal selection—two locations per line, one per free location—is affinely dependent; its circuit has no coincidence, at most two points per line, and no transversal triple, hence size ≥ 4 : again a contradiction. So $\dim V = 2r + f$ and $V = V(\pi)$. For sufficiency, in the block chart the functionals $\mathbf{1}_{B_j}$, $a\mathbf{1}_{B_j}$, $\mathbf{1}_C$ annihilate any dependency using ≤ 2 points per line block and ≤ 1 per free location (two locations of a line block carry distinct costs), so every circuit is a coincident pair or an in–block collinear triple, and Theorem 21.1 gives rank ≤ 1 on every spanning window. $\square$

A block with at most two distinct cost values is the *same* function space as its within–level refinement—affine in a through two values is no constraint—so the canonical presentation has two block types: *clusters*, within–level subsets (on this catalog singletons or the golden pair $\{\varphi, 1/\varphi\}$), and *line blocks* carrying at least three distinct cost values. Then $\dim V(\pi) = c + 2b_2$ with c clusters and b_2 line blocks.

Theorem 21.3 (Classification and counts). *The constant– $\frac{1}{4}$ k –families of the catalog are exactly the canonical $V(\pi)$ with $c + 2b_2 = k + 1$, and the counts are*

$$k = 2: 8, \quad k = 3: 56, \quad k = 4: 95, \quad k = 5: 31, \quad k = 6: 1.$$

*At $k = 2$ this recovers the eight surfaces of Theorem 12.3 in one line. At $k = 3$ the 26 double–indicator classes of Theorem 17.4 are joined by 30 split–affine classes $\langle \mathbf{1}, a, \mathbf{1}_B, a\mathbf{1}_B \rangle$ —the two–block partitions with at least three cost values on both sides; the five (3, 4)–splits whose 3–side is $\{\varphi, 1/\varphi, x\}$ coincide with $\langle \mathbf{1}, a, \mathbf{1}_{\{\varphi, 1/\varphi\}}, \mathbf{1}_x \rangle$ and are not new. At $k = 5$ the golden merge $\mathbb{R}[\log M]$ —the totally geodesic family of Theorem 16.3—appears among the 31; at $k = 6$ only the full simplex remains. [FORCED] (*t9_landscape.py*: exhaustive exact window verification of all $8 + 56 + 95 + 31$ families over $\mathbb{Q}(\log 2, \log 3, \log 5, \log \varphi)$; the five overlap equalities by symbolic rank; pairwise distinctness of the 56 certified at the true logarithms by 1540 interval rank–5 witnesses; twenty exact random controls violate flatness.)*

Corollary 21.4 (Open Problem 17.6 resolved; the conjecture refuted). Within–level indicator joins do *not* exhaust the landscape: a split–affine class contains no indicators beyond $\mathbf{1}_B$ and $\mathbf{1}_{B^c}$ (an affine 0/1 pattern across three or more distinct cost values is constant), so it is not an indicator join, yet it is constant– $\frac{1}{4}$. Theorem 17.4 stands exactly as the double–indicator stratum of the 56. [FORCED]

Remark 21.5 (The plane lemma, bypassed). The moment trichotomy replaces the $k = 2$ chain wholesale. The multiset failure of Remark 17.5 is not repaired but made irrelevant: a doubled

support point is a 2-circuit and never obstructs. At $k = 2$ the structure theorem *implies* the plane lemma (the plane holds no two skew lines, so $r \leq 1$ and $f \leq 1$), which is how the eight surfaces drop out.

Remark 21.6 (Scope: level multiplicities). The proofs use only that every cost level of the catalog holds at most two seeds; the theorem applies verbatim to the drop- K , drop- $1/\varphi$ and add- $\sqrt{7}$ catalogs. A catalog with a level of multiplicity ≥ 3 admits genuinely new *vertical* line blocks (a line inside one level, transverse to the cost coordinate), classified by the same proof with the line parameter replacing a ; the counts above are for the studied catalogs.

22 The coupled family curves

Remark 20.4 left two residuals: the geometry of the coupled family $(\text{cost}_1, \text{cost}_2, c)$ on the 49-cell tensor catalog, and whether Δ is determined by the charge data. Both close, the first with the machinery of §21.

Theorem 22.1 (The coupled family is not constant- $\frac{1}{4}$). *On the tensor catalog the triple $T = (\text{cost}_1, \text{cost}_2, c)$ spans, $\dim V = 4$, and its configuration contains a genuine 5-circuit—all five dependency weights nonzero, certified at the true logarithms by interval arithmetic—so by Theorem 21.2 it is not of partitioned-affine type, and the corresponding window matrix has rank ≥ 2 : the family is not windowwise flat. Directly, the Gauss obstructions at the uniform point are nonzero (rigorous enclosures), with sectional curvatures*

$$\sec(\text{cost}_1, \text{cost}_2) = \frac{1}{4} - 0.248531\dots \approx 0.0015, \quad \sec(\text{cost}_i, c) = \frac{1}{4} - 0.083802\dots \approx 0.1662.$$

The forced coupling of Theorem 20.1 is curvature-visible; the near-vanishing of the first plane is the ghost of the product geometry (Proposition 14.1). [FORCED]/[COMPUTED] (`t10_coupled.py`, 9/9; enclosures at 60 digits.)

Proposition 22.2 (Invisible to covariance, visible to curvature). *At the uniform point $\text{Cov}(\text{cost}_1, \text{cost}_2) = 0$ exactly—the marginals are independent there—yet the sectional curvatures above already deviate from $\frac{1}{4}$. The coupling enters through the compositum cost c , below the reach of the cross-covariance at that point: the covariance witness of Theorem 20.1 lives at $\beta \neq 0$, the curvature witness already at $\beta = 0$. [FORCED]*

Proposition 22.3 (Δ is not determined by the charge data). *The pairs $(\sqrt{2}, \sqrt{3})$ and $(\sqrt{2}, \sqrt{5})$ carry the same charge pair $(\mathbb{Z}/2, \mathbb{Z}/2)$, but*

$$\Delta(\sqrt{2}, \sqrt{3}) = \frac{5}{49} \log 2 - \frac{5}{49} \log 3 + \frac{20}{49} \log \varphi \neq \Delta(\sqrt{2}, \sqrt{5}) = \frac{5}{49} \log 2 + \frac{2}{49} \log 3 - \frac{1}{7} \log 5 + \frac{20}{49} \log \varphi,$$

exactly. The interaction tensor is a forced datum beyond the charge grading. [FORCED]

Remark 22.4 (The ledger closes). With §21 and this section, every enumerated open problem of the corpus is resolved. The temperature selection remains [DECLARED] by the dichotomy of Theorem 15.3—its honest final form—and the standing conditioning is, as throughout, the $\mathbb{Q}$ -linear independence of the seed logarithms.

23 Interaction with the trace form

Theorem 23.1 (Embedding obstruction). *The field’s flat trace–form geometry admits no totally geodesic isometric embedding into the surprisal sphere; any isometric surface realization is forced to satisfy $k_1 k_2 = -\frac{1}{4}$ (Gauss: $0 = \frac{1}{4} + k_1 k_2$), witnessed by the Clifford torus in $S^3(2)$ with principal curvatures $\pm\frac{1}{2}$. [FORCED]*

Theorem 23.2 (Fisher monotonicity). *Coarse–graining the field’s conjugate data into the catalog outcomes is a Markov morphism under which $I_{\text{surprisal}} \preceq I_{\text{trace}}$ (data–processing [4, 3]), equality iff sufficient. The catalog surprisal geometry is therefore a lossy, one–way contraction of the trace–form geometry. [FORCED]*

Together with Theorem 10.1 the picture is sharp: the arithmetic (trace–form) geometry is flat and signature–flexible; the surprisal geometry is curved ($\frac{1}{4}$) and positive definite; and the two–statistic field surface sits inside the latter as a constant–curvature– $\frac{1}{4}$ ruled surface—neither it nor the trace form is totally geodesic (Prop. 12.1), for the opposite reasons of wrong embedding and wrong intrinsic curvature respectively.

24 Epistemic ledger

25 Open problems

(1) *Resolved in this edition as a dichotomy* (Theorem 15.3): no invariance principle selects an operating point canonically—the labeled–endpoint isometry group is trivial, and the candidate principles (arc–length midpoint, maximal Fisher information, maximum entropy, peak heat capacity) select provably distinct temperatures. The [DECLARED] on the selection is the answer’s honest form, not a gap. (2) *Resolved in this edition* (§21): with necessity closed at every k (Theorem 19.2), constant $\frac{1}{4}$ is windowwise flatness, and the windowwise–flat spans are classified— $V = V(\pi)$, partitioned–affine, with catalog counts 8/56/95/31/1 for $k = 2, \dots, 6$. The indicator–join conjecture is refuted by the 30 split–affine classes at $k = 3$ (Corollary 21.4). (3) *Resolved in this edition*. The conceptual proof of the window identity is Theorem 13.4: $P = Z^2 \sum_{|s|=4} q_4(s) w_s$ by Sylvester’s determinant identity (pivot Z) plus Cauchy–Binet, with the heavy cofactor–1 law, the full coefficient law and 462–census, the factorization $q_4 = -\prod \Delta$, and the compound–ratio form of q as corollaries (Corollary 13.5). What was found by machine (140 exact divisions) is now a two–line determinantal mechanism; the collision/branch machinery is retired from the critical path (Remark 13.6). (4) *Resolved in this edition, on the first horn* (§20): the compositum’s own tensor catalog forces the coupling—the cost $\log M(p \otimes q)$ is provably non–separable, with interaction contrast exactly $\log 2$ (and $-6 \log \varphi$ in the golden sector), nonzero cross–Fisher covariance, and an interaction tensor of exact rank 5; the tensor charge law $\mathbb{Z}/\text{lcm}$ holds on all 28 pairs. The declared warp is promoted to a forced nonzero datum in the compositum reading. Its residuals are closed in §22: the coupled family is provably not constant– $\frac{1}{4}$ —the coupling is curvature–visible, already at the uniform point where the cross–covariance vanishes exactly—and Δ is not determined by the charge data. (5) *Resolved in this edition* (Theorem 19.2): constant $\frac{1}{4} \Rightarrow$ windowwise rank ≤ 1 at every k , by squarefree window collisions (the Salem square cannot act on windows) plus twin symmetry at the golden pair.

Outlook. What remains is outward–facing rather than internal: catalogs with cost levels of multiplicity ≥ 3 , where vertical line blocks enter the partitioned–affine classification (Remark 21.6); composita of two *distinct* catalogs, where the tensor construction and Δ generalize verbatim; the

coupled family’s curvature landscape away from the uniform point; and, as throughout, the standing conditioning of the collision analysis on the $\mathbb{Q}$ -linear independence of the seed logarithms.

A Reproducible verification

All load-bearing identities are checked in `field_surprisal.v2.py` (`sympy` exact arithmetic; `mpmath` for transcendental and numeric values), printing 12/12 passed. The checks: $Z = 17 + 4\sqrt{5}$, the affinity, and $I = \text{Var}(\log M)$ (§§3–4); the Bregman–KL identity and $A^* = -H$ (§5); $C = \beta^2 I$ and the peak-fluctuation temperature (§6); the vertex distance π and the information length (§7); the finite geometric closed form and the geometric variance of the keystone sub-model (§9); the two-statistic Fisher matrix and the peak-fluctuation temperature (§§6,10); and the embedding obstruction $k_1 k_2 = -\frac{1}{4}$ and the strict Fisher contraction (§11). Theorem 10.1—the exact constant curvature $\frac{1}{4}$ of the $(\log M, \deg)$ surface—is proved in the text and verified symbolically in `suspension_theorem.py` (9/9): the Brioschi curvature of a general indicator family equals $\frac{1}{4}$ identically in the free statistic, remains $\frac{1}{4}$ under arbitrary base measure (hence constant everywhere), the suspension metric has $g_{\psi\psi} = 1$, $g_{\theta\psi} = 0$ and curvature 1, the catalog specialization gives $\frac{1}{4}$, and generic controls give 4/21 and 857/16928. The Tier-2 results—the multi-statistic landscape (only the single-outcome indicator is constant $\frac{1}{4}$; the 3-set indicator and generic statistics vary) and the product geometry (block-diagonal Fisher, zero cross-curvature), and the charge grading (charge parity $= \mathbf{1}_{\{\varphi^4\}}$ is a second constant- $\frac{1}{4}$ surface; the charge-group order varies)—are verified in `field_surprisal_tier2.py` (15/15). The complete classification—no surface totally geodesic ($d \in \{6, 7\}$, $\log M^2 \notin V$), the ruled second fundamental form ($\Pi_{ac} = \Pi_{cc} = 0$, $\Pi_{aa} \neq 0$), the exactly eight constant- $\frac{1}{4}$ surfaces among all 127 indicators, and the ≈ 5000 -sample exclusion of non-indicators—is verified in `field_surprisal_classification.py` (9/9). Across the four harnesses (12/12, 9/9, 15/15, 9/9) a reviewer reruns every value; all numbered results are proved in full or established as marked.

The closures of §§13–16 are verified in six further harnesses, with exact `sympy` arithmetic over $\mathbb{Q}(\log 2, \log 3, \log 5, \log \varphi)$ at every decision boundary and floats display-only: `t1_core.py` (61/61: cost-value structure, the ruled system, the invariant-module dimensions $N(X)$, the $r = \dim V^2/V$ census, the Möbius elimination steps); `t1_reduction.py` (17/17: $Z^2 Dq = P$, $q = \text{Tr}(\Gamma N)$, the four-point rank-one law $q = \kappa q_4$, the rank-2 collision lattice); `t1_engine.py` (4/4: the full 462-monomial coefficient dictionary of P , support ≤ 3 vanishing, the window identity at all 140 heavy placements, and the ν -group census); `t1_branches.py` (16/16: window survival in every branch of the collision pencil, the merged six-point recursion with its rank-1 lattice, and identical window vanishing on the eight families in symbolic (α, β, γ)); `t2_temperature.py` (12/12: factorization rank tests, the lattice–Cauchy pathology, the exact anchors $Z(\mp 1)$, the identity $I' = -\kappa_3$, and the curve invariants); `t3_suspension.py` (24/24: the join metrics’ constant curvature for $k = 3, 4$ exactly, curvature routines validated on sphere, hyperbolic plane, flat plane, and the simplex, the catalog three-fold at $\frac{1}{4}$ to 10^{-18} , and the geodesic threshold). Session total: 134 checks, all exit 0.

Two harnesses added in this edition. `t1_windowproof.py` (44/44, exit 0) machine-proves Theorem 13.4 and Corollary 13.5: the Sylvester and Cauchy–Binet steps as fully symbolic polynomial identities at $m = 4, 5$; the factorization $q_4 = -\prod_{t \in s} \Delta_{s \setminus t}$ symbolically for four generic points; the compound-ratio form of q and the face law $q = \kappa q_4$ at exact rational instances for $m = 6, 7$; and, on the catalog ring $\mathbb{Q}[X_1, \dots, X_7, \log 2, \log 3, \log 5, \log \varphi]$, the dictionary-level equality of all 462 coefficients of P with $Z^2 \sum_s q_4(s) w_s$, string-exact against the artifact saved by `t1_engine.py`. *Ordering note:* that final cross-check reads the artifact `P_coeffs.pkl` written by `t1_engine.py`, so run `t1_engine.py` first for the full 44/44; standalone, the artifact check auto-skips (reported “not

counted”) and the tally is 43/43. The one-line runner `run_all.sh` executes all harnesses in a correct order. `t3c_partC_exact.py` (16/16, exit 0) re-proves the Part-C claims of `t3_suspension.py` (checks [016]–[024]) by explicit certificates that avoid symbolic-rank heuristics: the level-set annihilator $\prod_{r=1}^6 (a - v_r \mathbf{1}) = 0$ identically in $\mathbb{Q}[\log 2, \log 3, \log 5, \log \varphi]^7$, the explicit closure recursion for a^k ($k \geq 6$) with elementary-symmetric coefficients, the Vandermonde lower bound (conditioned, as everywhere, only on the $\mathbb{Q}$ -linear independence of the four logarithms), the golden-pair span equality, the $k \leq 4$ exclusion and $k = 5$ uniqueness, and the non-geodesy of the catalog three-fold.

Four harnesses close this edition’s remaining fronts. `t4_kwindows.py` proves the generalized window identity symbolically at $k = 3$, $m = 5$, checks it at exact rational instances $m = 6, 7$ with both Sylvester scalings, verifies the windowed Gauss-obstruction identity and the flat lemma, certifies the catalog three- and four-fold window matrices rank ≤ 1 over the coefficient field, and runs the double-indicator census by symbolic-rank membership (checks [001]–[013], exit-clean in-sandbox; the census block’s symbolic ranks exceed the execution ceiling there and are superseded by the second lane). `t4b_census_fast.py` (9/9, exit 0) is the census’s second lane: an independent block-system reduction deciding the same memberships by exact $\mathbb{Q}^5$ -annihilator and quadratic cross-ratio tests, completing all 7875 pairs (7812 nondegenerate, 312 flat, 26 classes) with sampled cross-lane agreement. `t5_catalog_census.py` (72/72, exit 0) proves the generic census theorem, the count formula, the collision lattices (ranks 2/1/1/2 with the named generators, $\mathbb{Z}$ -saturated), the sufficiency factorization, and the branch-rank necessity in every branch of all four catalogs, the full catalog reproducing exactly the eight-branch tree of `t1_branches.py`. `t6_selection.py` (18/18, exit 0) proves the exact anchors $Z(+1) = \frac{91}{30} - \frac{\sqrt{5}}{5}$ and $Z(-1) = 17 + 4\sqrt{5}$ in $\mathbb{Q}(\sqrt{5})$, the identity $I' = -\kappa_3$ symbolically, and the selection dichotomy’s numeric margins at dps 30/50 agreement below 10^{-12} .

Two harnesses close the final fronts. `t7_knecessity.py` (28/28, exit 0) is the machine lane of §19: the window-collision census of all four catalogs at every window size (every collision class is a single golden-swap pair; drop- τ has none), the cube fact for $\langle d_1, d_2 \rangle$, twin symmetry of the window functionals at $k = 2$ (all ten swap pairs and all ten twin-containing windows, by 4×4 determinants) and $k = 3$ (entrywise row agreement plus determinant spot-checks), the minor/rank equivalence, the $k = 3$ windowed identity $D_k G^{abcd} = Z^{k-2} \sum_s w_s [2 \times 2 \text{ minor}]$ at an exact rational instance ($m = 7$, all nine quadruples), and exact windowwise-rank recertification of the catalog three- and four-folds with a failing control. `t8_compositum.py` (13/13, exit 0) verifies §20: the 7×7 tensor-cost matrix as exact $\mathbb{Q}$ -vectors (the ten on-circle root products excluded symbolically; every remaining sign certified by rigorous interval arithmetic), the two exact interaction witnesses, rational-sector separability, the persistent golden tie, rank $\Delta = 5$ over $\mathbb{Q}(\log 2, \log 3, \log 5, \log \varphi)$, nonzero cross-Fisher covariance at $\beta \in \{1, -1, \sqrt{5}\}$, and the lcm charge law on all 28 pairs. A practical note: the exhaustive *symbolic* census lane of `t4_kwindows.py` (block 3) exceeds a sandbox execution ceiling; the fast $\mathbb{Q}^5$ -annihilator lane `t4b_census_fast.py` is the census’s practical lane, with sampled two-lane agreement.

Two final harnesses. `t9_landscape.py` (21/21, exit 0): the moment trichotomy symbolically at $d = 2, 3, 4$ and on exact instances; the full canonical enumeration with every window matrix certified rank ≤ 1 over $\mathbb{Q}(\log 2, \log 3, \log 5, \log \varphi)$ (26 + 30 classes at $k = 3$, 50 + 45 at $k = 4$, 30 + 1 at $k = 5$, the eight at $k = 2$); the five golden-overlap span equalities by symbolic rank; pairwise distinctness of the 56 at the true logarithms (1540 rank-5 witness minors, interval-certified); and twenty exact random controls violating flatness. `t10_coupled.py` (9/9, exit 0): the coupled triple’s spanning ($\dim V = 4$); a genuine 5-circuit and a rank- ≥ 2 window at the true logarithms; the three nonzero Gauss obstructions at the uniform point with certified sectional enclosures; the exact vanishing of the uniform cross-covariance; and the Δ -charge witness.

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claim

$Z = 17 + 4\sqrt{5}$; Mahler–Gibbs is an exponential family with statistic $\log M$
surprisal $= \beta \log M + \log Z$; Fisher $= \text{Var}_\beta(\log M)$; geometry $= S^{m-1}(2)$, $K = \frac{1}{4}$
dual flatness; KL = Bregman divergence of $\log Z$; $A^* = -H$
 $C(\beta) = \beta^2 I(\beta)$ (Fisher information is a heat capacity)
 $d_{\text{FR}} = 2 \arccos \text{BC}$; vertices mutually π apart
keystone sequence: shifted geometric law; $I = (\log \varphi)^2 r / (1 - r)^2$
embedding obstruction $k_1 k_2 = -\frac{1}{4}$; Fisher monotonicity (lossy contraction)
($\log M, \deg$) Fisher matrix SPD; peak–fluctuation β^* ; information length
no ($\log M, X$) surface is totally geodesic ($\log M$ spans a 6–dim algebra)
($\log M, \deg$) surface: constant curvature $\frac{1}{4}$, ruled (indicator–suspension)
among forced invariants only degree gives constant $\frac{1}{4}$; #outside/trace/house vary
product of independent catalogs: block–diagonal Fisher, zero cross–curvature
charge parity $= \mathbf{1}_{\{\varphi^4\}}$: a second constant– $\frac{1}{4}$ surface (dual to degree)
exactly eight constant– $\frac{1}{4}$ surfaces; non–indicator exclusion by branch calculus
master window identity $P = Z^2 \sum_s q_4(s) w_s$ (Sylvester + Cauchy–Binet); heavy cofactor 1
iterated join: multi–indicator families are round $\frac{1}{4}$ –spheres; catalog 3–fold
geodesic threshold $k = 5$, uniquely $\mathbb{R}[\log M]$ (the golden–merged simplex)
total information length $L_{\text{tot}} \approx 5.6462$; endpoints golden merge $\leftrightarrow \varphi^4$
coupling of two fields is a declared warp (Čencov), like β and c
Gibbs form $p \propto M^{-\beta}$ forced by product functoriality + measurable Cauchy
selection of a single operating temperature among the marked points
generalized window identity, every k (Sylvester + Cauchy–Binet over $(k+2)$ –subsets)
window criterion: rank ≤ 1 windows $\Rightarrow$ constant $\frac{1}{4}$; catalog 3– and 4–folds exact
 $k = 3$ double–indicator census: exactly 26 windowwise–flat classes (disjoint within–level pairs)
generic census: independent costs $\Rightarrow$ coefficientwise vanishing $\Rightarrow m$ surfaces
census count $\sum_\ell (2^{m_\ell} - 1)$; catalogs full/drop– K /drop– τ /add– $\sqrt{7}$: 8/7/6/9
branch–rank calculus (grouped window symbols, rank = #windows per branch)
selection dichotomy: no metric–canonical point; midpoint/max–Fisher/max–entropy/peak– C distinct
 k –necessity, every k : constant $\frac{1}{4} \Leftrightarrow$ windowwise rank ≤ 1 (squarefree collisions + twin symmetry)
window collisions are golden–only: the Salem square cannot act at window level
compositum forces coupling: contrast exactly $\log 2$; rational sector separable; golden tie persists; rank $\Delta = 5$
charge tensor law $n(p \otimes q) = \text{lcm}$ on all 28 compositum pairs; cross–Fisher $\text{Cov}_\beta \neq 0$
moment trichotomy: window circuits of size $2/3/\geq 4$ give rank $0/1/\geq 2$
landscape classified: constant– $\frac{1}{4} =$ partitioned–affine $V(\pi)$; counts 8/56/95/31/1, $k = 2..6$
indicator–join conjecture refuted: 30 split–affine classes $\langle 1, a, \mathbf{1}_B, a\mathbf{1}_B \rangle$
coupled family not constant– $\frac{1}{4}$: Gauss obstruction $\neq 0$ at uniform; Δ not charge–determined
 Z ’s integer 17; $I(1/\varphi) = \varphi^3$
